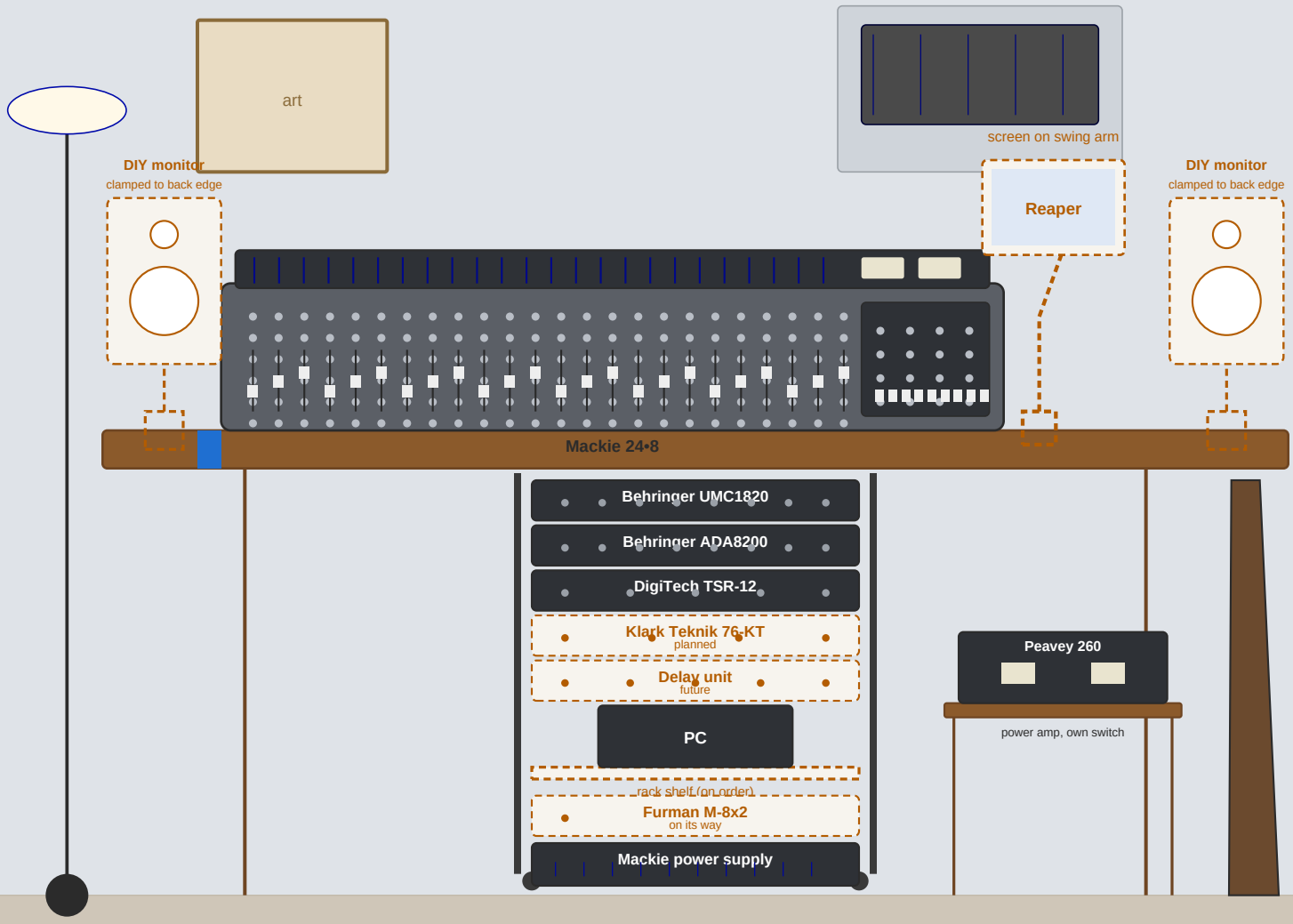
TOAD STUDIO

The Studio Book

Mackie 24•8 home studio, Champaign-Urbana • October 2026



What's drawn

In the studio now: Mackie 24•8 with meter bridge, UMC1820, ADA8200, TSR-12, Mackie power supply, the PC.

Planned: 76-KT, a delay unit, a rack shelf for the PC, Furman M-8x2, Peavey 260 on a side shelf, DIY monitors clamped to the back edge of the desk, and a computer screen on a swing arm clamped to the right of the console.

Monitor placement: tweeters at ear height, each speaker the same distance from your head as they are from each other. Keep the swing arm on the opposite side from the Peavey's power cord so the screen cable never crosses the snakes.

Josh Leitschuh • Contents on the next page

TOAD STUDIO: The Studio Book

Mackie 24•8 home studio, Champaign-Urbana. Revised October 15, 2026. Page numbers are at the bottom of every page; the Connection Diagram, Panning Guide and Goal Studio Diagram pages are landscape, so rotate them in your viewer if needed.

Contents

Studio Build Summary 4
Documents 4
The studio at a glance 5
Capacity at a glance 5
How channels get recorded 5
Aux sends on the 24•8 5
16-track playback and clock (UMC1820 + ADA8200) 5
Headphones and monitoring 6
Submixers and personal monitor mixes 6
Gain staging rules 6
Microphones 6
Electronic percussion 6
Rack additions and deals 7
Connecting more gear to the 24•8 7
Marshall MG250DFX 7
Effects and the Tame Impala sound 7
Analog tape: Tascam 246 (once repaired) 7
Garage PA 7
Power 8
Shopping status 8
Open items 8
Studio Setup Map 9
Rack order (top to bottom) 9
Console channel and panning map 9
Snake color order 10
UMC1820 and ADA8200 (Reaper inputs and playback 9-16) 11
Aux sends on the 24•8 11
PV6 submixer, electronic drums, drum machine and DJ 11
Mic notes, acoustic guitar and pedalboard 11
Tascam 246 analog tape (once repaired) 12
Capacity at a glance 12
Open connections and planned additions 12
Garage PA (STUDIO OUTPUT) 12
TSR-12 reverb 12
Wiring the back 12
Cables 13
Studio Connection Diagram 14
Signal flow 14
Power, instrument chains and snake colors 15
Monitoring: Outputs and Routes 16
Every output the console has 16
The MONITOR section, button by button 16
The PHONES sections (Phones 1 and Phones 2) 16
Talkback 17
Setups for common sessions 17
Rules that keep you out of trouble 18
Power: Start-Up and Shutdown 19
Where the Furman plugs in 19
What the Furman M-8x2 gives you 19
Which outlet gets what 19
Two things that stay off the Furman's switched outlets 20
Start-up, in order 20
Shutdown, in reverse 20
Details that matter 20
Gain Staging Guide 21
1. Zero everything first 21
2. Set each channel (one at a time) 21
3. Channel chart 21
4. What ends up in the recording 22
5. Outboard and other gear 22
6. Final checks 22
Setting Compression Levels 23
What the knobs do (76-KT) 23
Wiring and gain structure 23
Step by step 23
Starting points 23
If you buy the dbx alternative instead (266XS or 1066) 24
Common mistakes 24
EQ Settings, Channel by Channel 25
The EQ section on each 24•8 channel, top to bottom 25
Rules that matter more than any setting 25
Channels 1-6 26
Channels 7-12 26
Channels 13-18 27
Channels 19-24 27
Why each channel is set this way 28
Mic Placement 29
Rules that cover everything 29
Drums 29
Amps 30
Vocals, horn and acoustic 30
The band tracking together: bleed control 30
Room-dependent decisions (what your measurements would change) 31
Tascam 246: Setup and Recording 32
What's on the 246 (so the rest makes sense) 32
Connections 32
Levels: the one thing that matters most 32
246 channel settings 33
Recording a pass to tape, step by step 33
Using all four tracks 33

Contents, continued

Things that go wrong 34

Reaper Setup 35

1. Driver and audio device (once) 35
2. Name the inputs and outputs (once) 35
3. The track template (matches the console map) 35
4. Playback back to the console 36
5. Recording settings 36
6. MIDI for the e-drums and SamplePad 36
7. Mixing in Reaper: the plugins that replace outboard 36

Troubleshooting 37

FL Studio Setup 38

What's the same as Reaper (don't redo it) 38

1. Audio settings (once) 38
2. Map the 16 inputs to Mixer inserts (once, then save as template) 38
3. Playback back to the console 38
4. Recording a take 39
5. MIDI for the e-drums and SamplePad 39
6. Mixing in FL: the plugins that replace outboard 39

Reaper or FL Studio for this studio? 39

Troubleshooting 39

Pro Tools Setup 40

What's the same as Reaper (don't redo it) 40

1. Session and Playback Engine (once) 40
2. I/O Setup (once, then export the settings) 40
3. The session template 40
4. Recording 41
5. MIDI for the e-drums and SamplePad 41
6. Mixing: the plugins that replace outboard 41

Troubleshooting 41

GarageBand Setup 42

What's different about GarageBand 42

1. Preferences (once) 42
2. Make the tracks and set their inputs 42
3. The track layout 42
4. Recording 43
5. Playback back to the console 43
6. MIDI and the built-in instruments 43
7. Mixing: the plugins that replace outboard 43

When to move up 43

Troubleshooting 44

Panning Guide 45

Part 1: console pan knobs (tracking) 45

Part 2: final mix in Reaper 46

Alternate Setups 47

How to use this chapter 47

Alternate A: all direct outs 47

Alternate A channel map 47

Rewiring: what moves 48

Settings that change 48

Recording more than 16 at once 48

Main plan vs Alternate A 48

How the direct outs behave on the 24•8 49

Where the computer comes back in 49

Monitoring and latency 49

The noise test (do this before deciding) 49

Switching back 49

Alternate B: stereo playback (two outputs) 49

Alternate B wiring 50

Alternate B settings 50

16-track vs stereo playback 50

Moving to 16-track playback later 50

Alternate Setups: Connection Diagram 51

Studio Equipment List 52

Recording and mixing 52

Power 52

Monitoring 52

PA / powered mixers 52

Microphones and wireless 53

Amplifiers 53

Guitars 53

Keys, drums and electronics 53

Pedalboard 54

Studio Gear Needs List 55

Studio essentials 55

For the Tame Impala sound 55

Monitoring and recording extras 56

Only if needed 56

Compressor alternatives and gate option (not in the totals) 57

Totals 57

What's not on the list, and why 57

Goal Studio Diagram 59

Where every planned purchase plugs in 59

Studio Equipment Values 60

Console, rack and recording 60

Amplifiers 60

Guitars 60

Synths, drums and electronics 61

Microphones, wireless and headphones 61

Pedalboard 61

Powered PA mixers 62

Totals 62

How Each Piece Works 63

How it all connects 63

Recording 63

Monitoring 64

Microphones 64

Effects and processing 65

Other gear 66

Power-up and power-down 66

Quick gain-staging checklist 67

Studio Build Summary

Everything decided so far for the Mackie 24•8 studio, in one place. October 2026.

Documents

Section	What it's for
Studio Setup Map	Rack order, 24-channel map with pan settings, four snakes, ADA8200, TSR-12, PV6, Tascam 246, garage PA, wiring, cable list.
Studio Connection Diagram	Visual: signal flow (page 1), power and instrument chains with snake colors (page 2). Print and hang by the rack.
Monitoring: Outputs and Routes	Every output and what feeds it; the MONITOR and PHONES sections button by button; talkback; setups for each kind of session.
Power: Start-Up and Shutdown	What plugs into each Furman outlet, the switch-on order, the shutdown order, and why.
Gain Staging Guide	Zeroing the board, setting each of the 24 channels, outboard levels, troubleshooting.
Setting Compression Levels	Dialing in the 76-KT on a channel insert, with starting points per source and the dbx alternative settings.
EQ Settings, Channel by Channel	What each EQ knob does and what not to do with it; a knob-picture starting point for every channel.
Mic Placement	Distance, position and aim for every mic'd source; bleed control for band takes; which decisions depend on room size.
Tascam 246: Setup and Recording	The 246's own controls, how it connects, levels for tape, recording passes, bouncing, and four-track use.
Reaper Setup	ASIO driver, enabling all 16 inputs and outputs, the track template, playback routing, MIDI, recording and autosave settings.
FL Studio Setup	The same studio in FL Studio: Mixer insert inputs and Auto-map, the recording filter, outputs to the tape returns, MIDI.
Pro Tools Setup	The same studio in Pro Tools: Playback Engine, I/O Setup paths, session template, recording, MIDI. Needs Artist tier or above.
GarageBand Setup	The same studio on a Mac in GarageBand: an 8-input recorder with stereo-only playback; good for overdubs and demos.
Panning Guide	Console pan knobs for tracking (24 channels) and starting pan positions for the Reaper mix.
Alternate Setups	Other ways to wire the same studio, each with what changes and how to switch back. Alternate A: all direct outs, one source per track, with a noise test. Alternate B: stereo playback on two outputs, mixing in Reaper.
Studio Equipment List	Everything you own, October 2026.
Studio Gear Needs List	What's still missing, with price ranges, and what was ruled out.
Goal Studio Diagram	The prospective map: today's gear plus every planned purchase, wired in.
Studio Equipment Values	Rough used-market value of everything, with totals.
How Each Piece Works	What each piece of your gear does and how it connects to the 24•8; power order; quick gain checklist.

The studio at a glance

- Mackie 24•8 console (24 channels, 8 groups) on the desk; rolling rack underneath (about 12U): UMC1820, ADA8200, TSR-12, room for the 76-KT and a delay unit, a rack shelf for the computer, Furman M-8x2, Mackie power supply at the bottom. No rack headphone amp: personal mixers at the players' spots cover that. Planned around the desk: a screen on a swing arm right of the console, and the DIY monitors clamped to the back edge.
- 16 recording inputs in Reaper: groups 1-8 into the UMC1820 (Reaper 1-8) and 8 direct outs into the ADA8200 over ADAT (Reaper 9-16). 16 playback tracks: UMC1820 outputs to tape returns 1-8, ADA8200 outputs to tape returns 9-16. Reaper at 44.1 or 48 kHz.
- Four 8-channel snakes, same color order on all: brown, red, yellow, green, blue, gray, white, black. White tape = recording, black = playback 1-8, blue = direct outs, yellow = ADA8200 playback 9-16 (the 6 ft snake).
- All channels assigned except ch 22 (held for the drum machine if it has a stereo out).

Capacity at a glance

- **Console channels:** 23 of 24 used. Ch 22 is open, held for the drum machine if it has a stereo out; free for anything if it's mono.
- **Recording inputs:** 16 of 16 used. Groups 1-8 → UMC1820 (Reaper 1-8); direct outs → ADA8200 (Reaper 9-16). Anything new can be heard, but recording it means borrowing a group or direct-out path (like the Tascam on groups 7-8).
- **Playback:** 16 tracks. UMC1820 outputs 1-8 → tape returns 1-8; ADA8200 outputs → tape returns 9-16.
- **Still open for signals:** aux send 2 and AUX RETURN 2 (second effects unit); 2-TK and EXTERNAL inputs (monitor only: reference tracks, a phone); spare PV6 inputs (monitor-only sources); any mic input via the Radial ProDI once that channel's source is swapped.
- **If you outgrow 16 inputs:** the UMC1820's one ADAT input is taken by the ADA8200, so more recording channels means a bigger interface or a second one. Before that, submix monitor-only sources on the PV6.

How channels get recorded

- **Group channels** (drums, bass, guitars, Reface CS): press the channel's group button (1-2, 3-4, 5-6 or 7-8) and L/R. Pan picks the odd or even group, so pan settings stay fixed. Toms and rooms share groups 3-4; both guitar amps share group 6.
- **Direct-out channels** (lead vocal, wireless, trumpet, vocals 2-3, SH-201, drum machine): pan doesn't affect the recording. Press L/R only. The direct out is after the EQ, fader and mute, so keep faders near U (unity) and don't mute a channel while it's recording.
- **Monitor-only channels** (PV6 submix of e-drums, SamplePad 4 and DJ; Tascam 246 return): heard in the mix, not recorded through the console. E-drums and the SamplePad record as MIDI.
- On every recorded channel, the EQ and fader are part of what Reaper gets. Set them before a take and leave them alone during it.
- Lopsided headphones from hard-panned groups: press MONO or use a performer mix on AUX SEND 3/4. Neither changes the recording. Final panning happens in Reaper.

Aux sends on the 24•8

- **Aux 1 and 2:** each has its own knob on every channel. One PRE button covers both. The talkback mic can be sent into them.
- **Aux 3-6:** two knobs per channel. SHIFT makes them aux 3 and 4 (up) or aux 5 and 6 (down), so one channel can feed 3/4 or 5/6, not both. One PRE button covers the pair. SOURCE sends that channel's tape return (Reaper playback on MIX-B) instead of the live channel.
- **Headphones:** the two phones jacks can listen to aux 3/4 or aux 5/6 directly, not aux 1/2.
- **How this studio uses them:** effects on aux 1 and 2 with PRE up (TSR-12 reverb on 1, a delay on 2 later); headphone mixes on aux 3-6 with PRE down.
- **Setting up the aux 3/4 performer mix, step by step:** (1) on each channel the performer needs, SHIFT up and PRE down; (2) aux 3 knob = how loud that channel is in their left ear, aux 4 = right ear, so turn both up equally for a centered sound, or favor one for a little stereo; (3) raise the AUX SEND 3 and 4 masters in the master section to about noon; (4) on the Phones 2 section press AUX SEND 3/4 and bring its level up slowly; (5) keep Phones 1 on MONITOR so you hear the control room mix. Because PRE is down, moving channel faders won't change what the performer hears.
- **What's left to fill:** aux 2 is the only open effects send. Aux 5/6 is a second performer mix or two more mono cues for personal mixers. Aux returns 2, 3 and 4 are open stereo line inputs for a delay or any extra line source (DJ mixer, a phone) when the channels are full.

16-track playback and clock (UMC1820 + ADA8200)

- Two optical cables: ADA8200 ADAT out → UMC1820 optical in (recording), and UMC1820 ADAT out → ADA8200 ADAT in (playback). The ADA8200's 8 outputs feed tape returns 9-16 through the 6 ft snake.
- Clock with both cables: UMC1820 is the master; set the ADA8200's SYNC to ADAT IN until it shows LOCKED. With only the first cable, set the ADA8200 to its internal clock at Reaper's sample rate and set the UMC1820 to clock from its optical input.
- The ADA8200's outputs are XLR: use an XLR (female) to TRS snake, or 8 XLR-female-to-TRS-male adapters on a TRS snake. Worth checking before Todd sends the 6 ft snake.
- UMC1820 outputs 1-2 are its MAIN OUT jacks, which follow the front MAIN OUT knob. Turn that knob all the way up and leave it, so playback tracks 1-2 match 3-8.
- In Reaper, route tracks 9-16 to hardware outputs 11-18 (the ADAT outs); they play on MIX-B channels 9-16. The Reaper Setup page has the full template.

Headphones and monitoring

- The 24•8 has two headphone outputs, each with its own level and source (MONITOR, MIX-B, AUX 3/4, AUX 5/6, EXTERNAL).
- Basic: press MONITOR on both phones sections and L/R MIX in the monitor section. Reaper playback comes back on the tape returns: press MIX-B and raise MIX-B LEVEL. The full button-by-button walkthrough, every output, talkback and per-session setups are on the Monitoring page.
- Headphones: used Sony MDR-7506 and Sennheiser HD 280 Pro, both getting new pads.
- Headphone amp: decided against, to keep the rack space for a delay unit. Up to 2 people use Phones 1 and 2. For more, small mixers with headphone jacks at each player's spot, fed from aux sends 3-6; they double as personal headphone amps.
- Speakers: CONTROL ROOM out → Peavey 260 stereo power amp → DIY passive monitors (Parts Express kit). Until those are built, all mixing is on headphones.

Submixers and personal monitor mixes

- **Freeing console channels:** a small mixer can combine monitor-only sources (like the PV6 does for the e-drums, SamplePad 4 and DJ mixer) into one stereo pair. Don't submix anything you want on its own track; the blend gets locked in.
- **Musicians' own mixes:** feed a small mixer per musician from aux sends 3-6 with PRE down, each send carrying a group of instruments, e.g. aux 3 = drums, aux 4 = bass and guitars, aux 5 = vocals and trumpet, aux 6 = keys. Each player plugs headphones into their mixer and turns up "more me". This replaces using AUX 3/4 as a stereo pair for Phones 2.
- Because aux 3/4 and 5/6 share knobs, set SHIFT up on the drum, bass and guitar channels (aux 3 or 4) and SHIFT down on the vocal, trumpet and keys channels (aux 5 or 6).
- Any small mixer with a headphone jack works. The powered PA mixers aren't a good fit for this; use them for the garage PA and gigs.

Gain staging rules

- Set each 24•8 channel with its fader at U: press SOLO and raise the trim until peaks regularly reach 0 on the L/R meters. Solo on this console is after the fader, so the fader has to be at U first. Aim for Reaper peaks around -12 to -6 dBFS.
- **Mixer feeds what's recorded or the main mix** (PV6 submixer): set gains and master to unity once, then leave them. Small balance tweaks on its faders are fine.
- **Mixer only feeds someone's ears** (personal headphone mixers): musicians can change anything. It can't affect the recording.
- Tascam 246: tape meters around 0 VU, a little past for saturation.

Microphones

- Every owned mic has a spot: kick and snare mics, room mics, SM58-style (3) on toms and guitar amp 2, SM57 on guitar amp 1, SM58 on trumpet, Worker Bee on lead vocal, Peavey on vocal 2, Turner 95D on vocal 3. Bass goes through its DI out.
- No rack mic preamp needed: the 24•8 has 24 preamps and the ADA8200 adds 8. Next mic to buy, if any: another SM57.
- Phantom power is switched in blocks of eight channels (1-8, 9-16, 17-24). The Worker Bee on ch 10 means channels 9-16 all get 48V. That's fine for balanced dynamic mics. Check the wireless receiver's manual, and see the Turner note below.
- Turner 95D: made in high- and low-impedance versions with Amphenol MC1M or MC2M connectors; check the OHMS box. High: needs a matching transformer. Low (200 ohms): needs an adapter or rewired cable to XLR. Until its wiring is confirmed, don't plug it into a block with phantom on. Output is low (about 1.25 mV/Pa) and range narrow (100 Hz-10 kHz): a lo-fi character mic, great for a vintage vocal or a crushed mono room mic.
- Inline mic booster (Cloudlifter CL-1, Triton FetHead, sE DM1): only if a quiet vocal needs the trim near max and you hear hiss. Dynamic mics only, never condensers.
- Acoustic guitar overdubs: Worker Bee on ch 10, aimed at the 12th fret, 8-12 inches away. Keep it off the PV6, which isn't recorded.

Electronic percussion

- Roland e-drum pad and Alesis SamplePad 4 (5 pads total) go into the PV6 for monitoring and record as MIDI through the UMC1820's MIDI in or USB. One pad can play any sound; build parts a drum at a time.

Rack additions and deals

- Priority: (1) a compressor; (2) a patchbay; (3) a Pultec-style EQ such as the Klark Teknik EQP-KT, not a graphic EQ; (4) a rack drawer for patch cables.
- Compressor decision: the Klark Teknik 76-KT (1176-style) is the primary choice and is in the totals. Alternatives if one doesn't turn up, or for four channels later: two dbx 266XS from Drew's listing ($110 the pair, clean, with gates), or two dbx 1066 from Cory's listing ($150 each, better built, same four channels).
- Cory Pollard's listing also has 2 patchbays ($20 each) and 2 Hosa 8-channel TS snakes ($20 each): $80 for the four, worth taking even without the 1066s. Test every patchbay jack and each snake tail before paying.
- Klark Teknik DN514+ quad gate, $100 with free shipping: a fair price. It goes on the inserts of kick, snare and both toms and tightens the drums before they're recorded. Reaper can gate for free afterwards, so buy it only if tom ring and bleed are bothering you now. It takes 1U; with the headphone amp dropped, the rack has room for it under the delay unit.
- Klark Teknik 76-KT: the one in that listing sold. Still on the needs list as the 1176-style character compressor; if you buy the 1066s it becomes a nice-to-have at around $100-135 used.
- Skip from that listing: Warm Audio WA273-EQ ($1,100, a luxury), dbx 1074 quad gate (Reaper can gate later), dbx 231 graphic EQ (PA use only).
- Behringer HA8000 headphone amp listed at $50: passed on it; the rack slot goes to a delay unit and personal mixers handle extra headphones.
- Power amp: covered by the Peavey 260 (vintage 1970s stereo power amp). It drives passive speakers only; it isn't rack-mount, so it goes on a shelf.

Connecting more gear to the 24•8

- Studio monitors: the biggest gap. Plan: CONTROL ROOM out → Peavey 260 → DIY passive monitors built from a Parts Express kit (C-Note pair about $160). Get the Peavey checked by a tech, match speaker impedance to its rating, use real speaker cable, power it on last. JBL 305P MkII powered monitors remain the plug-and-play backup.
- A delay on AUX SEND 2 / AUX RETURN 2 (aux 1 is the TSR-12).
- Personal headphone mixers on AUX SENDS 3-6.
- Radial ProDI into any mic input: records a clean guitar or bass track alongside the amp mic, so it can be re-amped later; also for an acoustic pickup. Synths, drum machines and the DJ mixer don't need one.
- Reference tracks into the 2-TK input to compare mixes against Tame Impala songs.

Marshall MG250DFX

- The rear jacks are FX send, FX return(s), footswitch and CD input. The jack that didn't work was most likely the FX return: it skips the preamp, so a guitar plugged in there sounds weak or silent, and anything in it cuts off the preamp. A photo of the back panel will confirm.
- Use it as a loop: drive and fuzz pedals in front of the amp; delay, reverb, modulation and the 720 looper between FX SEND and FX RETURN.

Effects and the Tame Impala sound

- TSR-12 on aux send 1 (PRE up), returned on aux return 1, mix 100% wet. Monitor-only; Reaper records dry.
- Klark Teknik 76-KT (1176-style FET) on a channel insert via Y cable for squashed drums and vocals. Mackie inserts: tip = send, ring = return.
- Guitar chain: guitar → pedalboard (Big Muff, DBA Exploding Head, Cock Fight Plus, JHS gain, La Grotte) → Diamond Vibrato → MXR Phase 90 → 720 looper → amp. Vibrato needs a 9-18V supply; Phase 90 a 9V adapter or battery.
- Kevin Parker notes: SM57-style dynamics on drums and amps, heavy 70s drum compression and saturation, Juno-106 run through guitar pedals (run the SH-201 through the pedalboard into ch 17), Vox AC30, SSL-style bus compressor, Manley Vari-Mu on the master. Reaper's ReaComp can cover the mix bus.
- Tascam 246 tape saturation is another route to the 70s drum sound once it's repaired.

Analog tape: Tascam 246 (once repaired)

- To tape: 24•8 MAIN MIX out L/R → two 246 line inputs. From tape: 246 stereo line out → ch 23-24.
- To capture tape into Reaper, assign ch 23-24 to groups 7-8 in place of the Reface. Not synced: line takes up by ear or with a click.
- Full setup, the 246's own mixer settings, dbx, tape speed, bouncing and punch-ins are on the Tascam 246 page.

Garage PA

- STUDIO OUTPUT L/R → one powered mixer (Powerpod 410 or EMX62M) in the garage for playback or a monitor mix. Talkback reaches it through PHONES & STUDIO.
- The other powered mixer can take the MAIN MIX outs for live shows. Garage mics would need a long XLR snake back to the console.

Power

- The Furman plugs straight into a grounded wall outlet (never a power strip). All rack audio gear plugs into its eight switched rear outlets (won on eBay, on its way). Two things stay off it: the Peavey 260, on its own wall outlet so it can be last on and first off, and the computer, so a tripped breaker can't kill a session. Keep the Mackie power supply's umbilical away from the snakes.
- Start-up: levels down → Furman switch (console PSU, UMC1820, ADA8200, outboard all come up) → computer and Reaper → wait 20-30 s → Peavey 260 → bring levels up. Shutdown in reverse: levels down, Peavey off, computer down, Furman off. Full page: Power: Start-Up and Shutdown.
- If the window AC causes clicks or hum, power the studio from another circuit.

Shopping status

Needs list	Low	High
Studio essentials	$163	$268
Essentials + Tame Impala sound	$667	$896
Essentials + Tame Impala + monitoring extras	$1,067	$1,657
Everything, including only-if-needed items	$1,172	$1,947

Details and what each tier includes are in the Studio Gear Needs List.

Open items

- Tascam 246 repair (won't play tape).
- Run the group-vs-direct-out noise test (Alternate Setups) and decide whether to try Alternate A.
- Used MDR-7506 and HD 280 Pro: order pads. Confirm the HD 280 Pro is a 2016 or newer unit before buying the genuine Sennheiser pad kit.
- Power supplies for the Diamond Vibrato (9-18V) and Phase 90 (9V).
- Identify the 1970s Roland drum machine model (decides whether it needs ch 22 for a stereo out, and narrows its value).
- Check the Turner 95D's impedance and wiring before plugging it into the console.
- Send photos when back in the studio: Turner OHMS box and connector, back of the console, Marshall back panel, second pedalboard.
- Measure the live room and control room (length, width, ceiling height; which walls are hard or soft) so the Mic Placement page's room-dependent items become exact.
- Find a used 76-KT (around $100-135). Decide on Cory's patchbays and snakes, and whether the DN514+ gate is worth it now.
- Swap the 3 ft snake with Todd for the 6 ft one (the 3 ft doesn't reach); ideally XLR (female) to TRS for the ADA8200 outputs. Until then, play back in stereo (Alternate B).
- Third snake, XLR cables, second optical cable, MIDI cable, Tascam and DJ cables (see the Setup Map cable list).
- Install the Furman when it arrives: just above the Mackie power supply with one empty space between.
- Peavey 260: tech check, then build the monitors.

Studio Setup Map

Rack order, channel map, four snakes, effects routing and cables for the Mackie 24•8 studio. October 2026.

Rack order (top to bottom)

Position	Unit	Status	Why it goes here
1 (top)	Behringer UMC1820	In the rack	Used most: gain knobs, front inputs 1-2, meters. Closest to the desk.
2	Behringer ADA8200	In the rack	Direct-out channels record here (Reaper inputs 9-16); its outputs play Reaper tracks 9-16 to tape returns 9-16. Short optical cables to the UMC1820.
3	DigiTech TSR-12	In the rack	Reverb/delay on aux send 1.
4	Compressor: Klark Teknik 76-KT or dbx 1066	Not bought yet	Adjusted often while tracking.
5	Delay / second effects unit	Future	On aux send 2 / aux return 2. Headphones for 3+ people are handled by personal mixers at the players' spots, not a rack headphone amp.
6	Rack shelf with the computer	Shelf on order	A 2U vented shelf. The PC's own fan pulls air through the gap, which also keeps power-supply heat off the gear above. Leave the front of the shelf clear.
7	Furman M-8x2	On its way	Everything plugs in here; one switch handles power.
8 (bottom)	Mackie 24•8 power supply	In the rack	Heaviest, runs warm, can put hum into nearby audio gear. Farthest from the UMC1820 and TSR-12.

The Peavey 260 power amp isn't rack-mount: it sits on a shelf near the rack with air around it.

Console channel and panning map

Groups 1-8 feed UMC1820 inputs 1-8 (Reaper 1-8). Direct outs feed ADA8200 inputs 1-8 (Reaper 9-16). Channel 22 is open.

Ch	Source	Cable	Pan	Records to (Reaper input)
1	Kick	XLR	Hard L	Group 1 (Reaper 1)
2	Snare	XLR	Hard R	Group 2 (Reaper 2)
3	Rack tom (SM58-style)	XLR	10 o'clock	Groups 3-4 (Reaper 3-4)
4	Floor tom (SM58-style)	XLR	2 o'clock	Groups 3-4 (Reaper 3-4)
5	Room L	XLR	Hard L	Groups 3-4 (Reaper 3-4)
6	Room R	XLR	Hard R	Groups 3-4 (Reaper 3-4)
7	Bass amp (DI out or mic)	XLR	Hard L	Group 5 (Reaper 5)
8	Magnatone (SM57)	XLR	Hard R	Group 6 (Reaper 6)
9	Marshall or Orange (SM58-style)	XLR	Hard R	Group 6 (Reaper 6)
10	Lead vocal (Worker Bee)	XLR	Center	Direct out → ADA8200 in 1 (Reaper 9)
11	ATW-R3210 wireless	XLR	Center	Direct out → ADA8200 in 2 (Reaper 10)
12	Reface CS L	1/4" TS	Hard L	Group 7 (Reaper 7)
13	Reface CS R	1/4" TS	Hard R	Group 8 (Reaper 8)
14	Trumpet (SM58)	XLR	1 o'clock	Direct out → ADA8200 in 3 (Reaper 11)
15	Vocal 2 (Peavey)	XLR	10 o'clock	Direct out → ADA8200 in 4 (Reaper 12)
16	Vocal 3 (Turner 95D)	XLR	2 o'clock	Direct out → ADA8200 in 5 (Reaper 13)

Ch	Source	Cable	Pan	Records to (Reaper input)
17	Roland SH-201 L	1/4" TS	9 o'clock	Direct out → ADA8200 in 6 (Reaper 14)
18	Roland SH-201 R	1/4" TS	3 o'clock	Direct out → ADA8200 in 7 (Reaper 15)
19	Peavey PV6 main out L	1/4"	Hard L	Monitor only (e-drums, SamplePad 4, DJ)
20	Peavey PV6 main out R	1/4"	Hard R	Monitor only (e-drums, SamplePad 4, DJ)
21	1970s Roland drum machine	1/4" TS	Center	Direct out → ADA8200 in 8 (Reaper 16)
22	Open (drum machine R if stereo)	—	—	—
23	Tascam 246 line out L	RCA to 1/4"	Hard L	Monitor; assign to groups 7-8 to capture into Reaper
24	Tascam 246 line out R	RCA to 1/4"	Hard R	Monitor; assign to groups 7-8 to capture into Reaper

Group channels (1-9, 12-13): press the group button for that pair (1-2, 3-4, 5-6 or 7-8) plus L/R to hear it. Pan picks the group, so keep these settings fixed. If they sound lopsided in the headphones, press MONO or use the AUX SEND 3/4 performer mix; neither changes the recording. **Direct-out channels** (10-11, 14-18, 21): pan doesn't affect what's recorded, so set it for the headphones. Press L/R only, no group buttons. The direct out is after the EQ, fader and mute: keep faders near U (unity) and don't mute while recording. **Monitor-only channels** (19-20 and 23-24): press L/R only; they aren't recorded through the console. **Other layouts:** an all-direct-outs version of this map (every channel 1-16 on its own track, no groups) is in the Alternate Setups chapter, with how to switch.

Mics: every mic you own has a spot. Kick and snare mics on 1-2, room mics on 5-6, bass through its DI out. If the Turner moves to a mono room "crush" mic, vocal 3 needs another mic; an SM57 is the best add. **Phantom power** is switched in blocks of eight channels (1-8, 9-16, 17-24): the Worker Bee on ch 10 puts 48V on the mic inputs of 9-16.

Snake color order

Same order on both ends of all four snakes. UMC1820 inputs 1-2 are on the front panel; inputs 3-8 are on the back.

#	Color	Recording (white tape)	Playback 1-8 (black tape)	Direct outs (blue tape)	Playback 9-16 (yellow tape)
1	Brown	Group out 1 → UMC1820 in 1 (front)	UMC1820 out 1 → tape return 1 (MAIN OUT)	Ch 10 direct out → ADA8200 in 1	ADA8200 out 1 → tape return 9
2	Red	Group out 2 → UMC1820 in 2 (front)	UMC1820 out 2 → tape return 2 (MAIN OUT)	Ch 11 direct out → ADA8200 in 2	ADA8200 out 2 → tape return 10
3	Yellow	Group out 3 → UMC1820 in 3	UMC1820 out 3 → tape return 3	Ch 14 direct out → ADA8200 in 3	ADA8200 out 3 → tape return 11
4	Green	Group out 4 → UMC1820 in 4	UMC1820 out 4 → tape return 4	Ch 15 direct out → ADA8200 in 4	ADA8200 out 4 → tape return 12
5	Blue	Group out 5 → UMC1820 in 5	UMC1820 out 5 → tape return 5	Ch 16 direct out → ADA8200 in 5	ADA8200 out 5 → tape return 13
6	Gray	Group out 6 → UMC1820 in 6	UMC1820 out 6 → tape return 6	Ch 17 direct out → ADA8200 in 6	ADA8200 out 6 → tape return 14
7	White	Group out 7 → UMC1820 in 7	UMC1820 out 7 → tape return 7	Ch 18 direct out → ADA8200 in 7	ADA8200 out 7 → tape return 15
8	Black	Group out 8 → UMC1820 in 8	UMC1820 out 8 → tape return 8	Ch 21 direct out → ADA8200 in 8	ADA8200 out 8 → tape return 16

Check each color one at a time: brown into group out 1, confirm signal on input 1 in Reaper, then continue through black. UMC1820 outputs 1-2 are its MAIN OUT jacks, which follow the front MAIN OUT knob: turn that knob all the way up and leave it so playback tracks 1-2 match 3-8.

UMC1820 and ADA8200 (Reaper inputs and playback 9-16)

- Recording: optical cable from the ADA8200's ADAT out to the UMC1820's optical in. Its line inputs 1-8 (on the front) take 24•8 direct outs from channels 10, 11, 14-18 and 21 through the third snake (blue tape).
- 16-track playback: second optical cable from the UMC1820's ADAT out to the ADA8200's ADAT in. The ADA8200's 8 outputs (XLR, on the back) feed tape returns 9-16 through the 6 ft snake (yellow tape). In Reaper, route tracks 9-16 to the 8 ADAT outputs; they play back on MIX-B channels 9-16.
- The ADA8200's outputs are XLR: use an XLR (female) to TRS snake, or 8 XLR-female-to-TRS-male adapters on the rack end of a TRS snake.
- Clock, with both optical cables connected: the UMC1820 is the master; set the ADA8200's SYNC switch to ADAT IN until it shows LOCKED.
- Clock, with only the ADA8200-to-UMC1820 cable: set the ADA8200 to its internal clock at the same sample rate as Reaper, and set the UMC1820's clock source to its optical input. Clicks and pops on inputs 9-16 mean the clock isn't set right.
- Keep Reaper at 44.1 or 48 kHz. At 88.2/96 kHz, ADAT carries only 4 channels.

Aux sends on the 24•8

- Aux 1 and 2 have their own knobs on every channel and share one PRE button. Talkback can be sent into them.
- Aux 3-6 share two knobs: SHIFT up = aux 3 and 4, SHIFT down = aux 5 and 6, so one channel feeds 3/4 or 5/6, not both. One PRE button covers the pair. SOURCE sends the channel's tape return (MIX-B) instead of the live channel.
- The phones jacks can listen to aux 3/4 or 5/6, not 1/2. So: effects on aux 1 and 2 (PRE up), headphone mixes on aux 3-6 (PRE down).
- Aux 3/4 performer mix: on each channel the performer needs, SHIFT up, PRE down, aux 3 = left ear, aux 4 = right ear (equal for centered). AUX SEND 3 and 4 masters to about noon. On Phones 2 press AUX SEND 3/4 and raise its level. Phones 1 stays on MONITOR for you.
- Still open: aux 2 (second effect), aux 5/6 (a second performer mix or two mono cues), aux returns 2-4 (stereo line inputs for a delay or any extra source).

PV6 submixer, electronic drums, drum machine and DJ

- Peavey PV6 as a monitor submixer: e-drum module, SamplePad 4 and DJ mixer into the PV6; PV6 main out into ch 19-20, panned L/R, L/R button only. Set the balance between them on the PV6 and the overall level on the console.
- Keep the PV6's channel gains and master at unity (normal) settings to avoid adding hiss. Only monitor-only sources go through it; anything you want on its own track goes straight into the 24•8.
- E-drums: record the pad as MIDI. Module MIDI out to the UMC1820's MIDI in, then record a MIDI track in Reaper. One pad can play any sound, so build parts one drum at a time and replace sounds later.
- Alesis SamplePad 4: main output into the PV6 with the e-drums. Record it as MIDI through its USB port to the computer. The UMC1820 has one MIDI in, which the e-drum module uses.
- 1970s Roland drum machine: output to ch 21, recorded through its direct out on ADA8200 in 8 (Reaper 16). Use ch 22 if it has a stereo out.
- DJ sets: record on the DJ mixer's record out, or temporarily repatch the SH-201's ADA8200 inputs 6-7.

Mic notes, acoustic guitar and pedalboard

- Turner 95D: check the OHMS box on the label. High impedance: it needs an inline Hi-Z to low-Z matching transformer before the console's XLR input. Low (200 ohms): it needs an adapter or rewired cable from its Amphenol connector to XLR. Until that's sorted, keep it off channels 9-16 while the Worker Bee's phantom power is on. Great as a lo-fi vocal mic or a mono room mic crushed through a compressor.
- Acoustic guitar (overdubs): Worker Bee on ch 10, aimed at the 12th fret about 8-12 inches away. Ch 10 records through its direct out. Keep it off the PV6, which isn't recorded.
- Pedalboard (720 looper, La Grotte, DBA Exploding Head, Big Muff, Cock Fight Plus, JHS gain): guitar → drive and fuzz pedals → Diamond Vibrato → Phase 90 → 720 looper → amp. The Big Muff covers the fuzz role.
- Marshall MG250DFX FX loop: drive and fuzz pedals go in front of the amp's front input; delay, reverb, modulation and the 720 looper can go in the loop instead (FX SEND → pedals → FX RETURN). The FX RETURN feeds the power amp directly: a guitar plugged straight in there sounds weak or silent, and anything in the return cuts off the preamp.
- Synths through pedals, Kevin Parker style: SH-201 L out → pedalboard → ch 17. Keep the synth's volume low going in; synths run much hotter than guitars.

Tascam 246 analog tape (once repaired)

- To tape: the 24•8's MAIN MIX out L/R feeds two 246 line inputs. Whatever is in the L/R mix (a Reaper playback, a stem, or a live take) goes to the cassette tracks you arm on the 246.
- From tape: the 246's stereo line out feeds ch 23-24. Balance its 4 tracks on the 246's own mixer; set overall level on the console. Keep ch 23-24 out of the L/R mix while recording to tape, or the tape output feeds back into its own input.
- Levels: the 24•8 runs hotter than the 246. Start with the 246's input trims low and raise them until tape meters peak around 0 VU; push slightly past for more tape saturation.
- Capturing tape into Reaper: assign ch 23-24 to groups 7-8 (temporarily in place of the Reface CS) and record Reaper inputs 7-8. The cassette isn't synced to Reaper, so line takes up by ear or with a click at the start.
- Good uses: run drums or a full mix through tape for 70s-style saturation, or record 4-track ideas on the 246 by itself (its own mic inputs) and dump them in later.

Capacity at a glance

- **Console channels:** 23 of 24 used. Ch 22 is open, held for the drum machine if it has a stereo out; free for anything if it's mono.
- **Recording inputs:** 16 of 16 used. Groups 1-8 → UMC1820 (Reaper 1-8); direct outs → ADA8200 (Reaper 9-16). Anything new can be heard, but recording it means borrowing a group or direct-out path (like the Tascam on groups 7-8).
- **Playback:** 16 tracks. UMC1820 outputs 1-8 → tape returns 1-8; ADA8200 outputs → tape returns 9-16.
- **Still open for signals:** aux send 2 and AUX RETURN 2 (second effects unit); 2-TK and EXTERNAL inputs (monitor only: reference tracks, a phone); spare PV6 inputs (monitor-only sources); any mic input via the Radial ProDI once that channel's source is swapped.
- **If you outgrow 16 inputs:** the UMC1820's one ADAT input is taken by the ADA8200, so more recording channels means a bigger interface or a second one. Before that, submix monitor-only sources on the PV6.

Open connections and planned additions

- Monitors: CONTROL ROOM outputs L/R → Peavey 260 inputs (channel A = left, B = right; TRS-to-TS cables) → speaker cable → DIY passive monitors (Parts Express kit).
- Peavey 260 first use: tech check first. Speaker impedance at or above the amp's rated minimum (back panel). Level knobs low, CONTROL ROOM level low, then bring both up while watching the VU meters. Power it on last and off first.
- AUX SEND 2 → a delay (second effects unit, or a pedal with the send kept low) → AUX RETURN 2.
- Channel inserts → compressor (76-KT, or a pair of dbx 266XS / 1066) with insert Y cables: drum pair, lead vocal, bass.
- Channel inserts 1-4 → Klark Teknik DN514+ quad gate (option) with 4 insert Y cables: kick, snare, rack tom, floor tom. Set thresholds with the drummer at full volume; use the LF/HF key filters so cymbals don't open the tom gates.
- AUX SENDS 3-6 → small personal headphone mixers, one per musician (aux 3 drums, 4 bass and guitars, 5 vocals and trumpet, 6 keys), or AUX SEND 3/4 → an 8-channel headphone amp.
- Any mic input: Radial ProDI for a clean guitar or bass track for re-amping, or an acoustic pickup.
- 2-TK input: phone or computer (3.5mm to RCA) for reference tracks; press 2-TK in the monitor section.
- Patchbay in the rack for inserts and repatching (groups 7-8 swaps, ADA8200 borrowing).
- Garage alternative: the Peavey 260 can also drive passive PA speakers from STUDIO OUTPUT, freeing both powered mixers for gigs.

Garage PA (STUDIO OUTPUT)

- Feed one powered mixer (Phonic Powerpod 410 or Yamaha EMX62M) from the 24•8's STUDIO OUTPUT L/R. Choose its source in the MONITOR section and set level with the STUDIO knob.
- Use it for playback or a monitor mix in the garage. TALKBACK → PHONES & STUDIO lets you talk to whoever is out there.
- For live shows, the other powered mixer can take the MAIN MIX outputs instead.
- Garage mics would need their own long XLR snake back to the console; not included here.

TSR-12 reverb

- Wiring: 24•8 AUX SEND 1 → TSR-12 left input; TSR-12 L/R outputs → AUX RETURN 1 L/R.
- TSR-12: mix 100% wet; set input level so the meter peaks just under overload.
- Per channel: aux 1 to about noon, PRE up (not pressed). Master AUX SEND 1 to about noon; raise AUX RETURN 1 and assign it to L/R.
- Don't press TALKBACK → AUX SEND 1 while tracking. Reverb is monitor-only; Reaper records dry.

Wiring the back

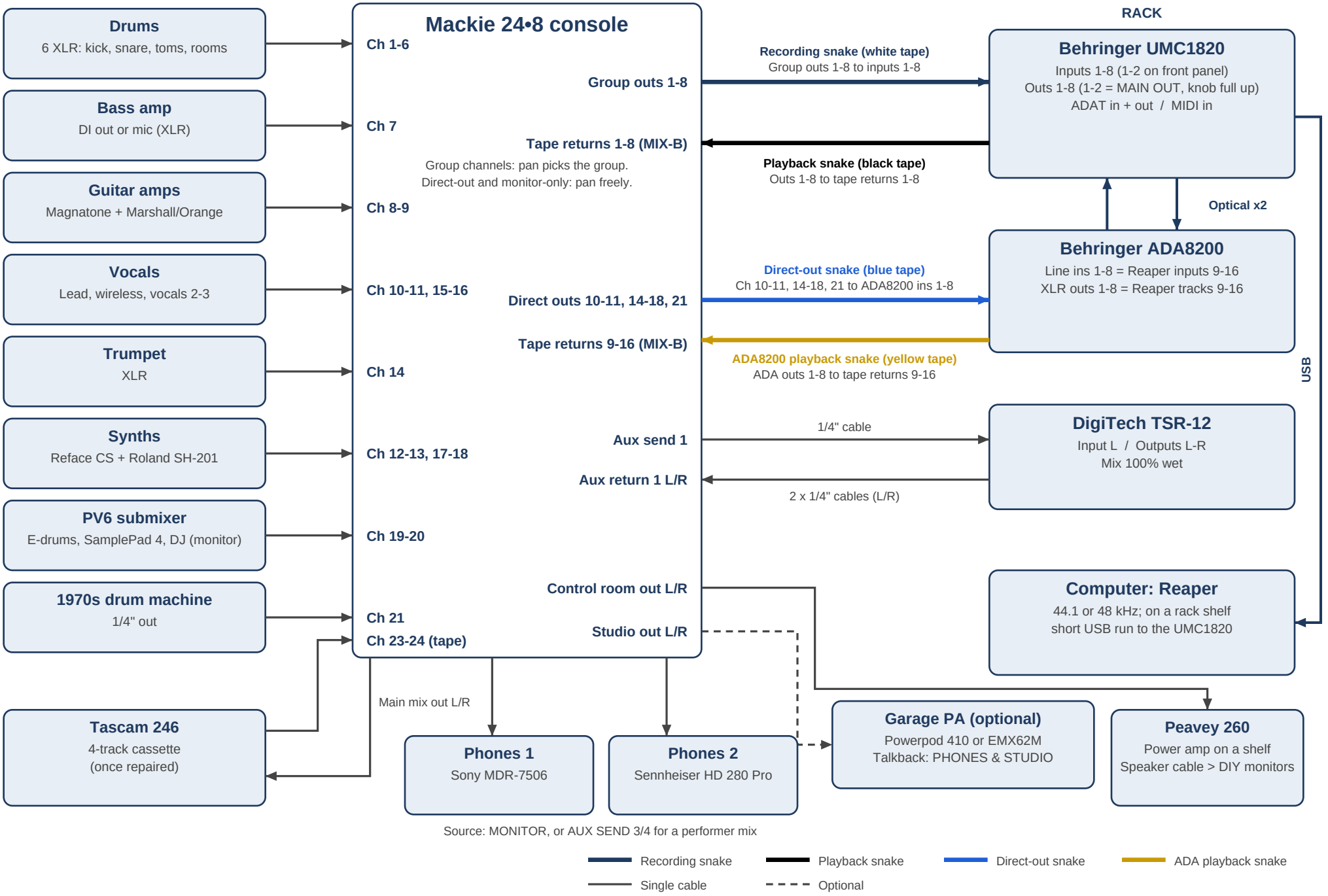
- Power cords down one rear rail, audio cables down the other. Snakes through the wire basket.
- Keep the power supply's umbilical away from the snakes and the coiled snake slack; cross only at a right angle.
- Plug the power supply, UMC1820, ADA8200 and outboard gear into the Furman.
- The computer lives on a rack shelf below the headphone-amp space, so the USB run to the UMC1820 is short. Its power cord goes to the wall, not the Furman.
- If the window AC shares the wall circuit and causes clicks or hum, power the studio from another circuit.

Cables

Cable	Qty	Use
8-channel TRS snake, 20 ft	2 owned + 1 needed	Recording, playback 1-8, and direct outs to the ADA8200 (all 8 tails used). Eight single TRS cables also work for the third.
8-channel snake, 6 ft	1 (swapping the 3 ft)	ADA8200 outputs → tape returns 9-16. XLR (female) to TRS, or a TRS snake plus 8 XLR-to-TRS adapters. Not needed for stereo playback (Alternate Setups, Alternate B).
XLR mic cable	14 + 1 spare	15-20 ft for drums, amps and trumpet; 25 ft for vocals and room mics.
1/4" TS line cable	14	Reface CS to ch 12-13 (2). SH-201 to ch 17-18 (2). E-drum module to PV6 (2). SamplePad 4 to PV6 (2). PV6 main out to ch 19-20 (2). Drum machine to ch 21 (1). TSR-12: aux send 1 to input, L/R outs to aux return 1 (3, short).
Optical (TOSLINK) cable	2	ADA8200 ADAT out → UMC1820 optical in; UMC1820 ADAT out → ADA8200 ADAT in.
MIDI cable (5-pin DIN)	1	E-drum module MIDI out → UMC1820 MIDI in.
Insert Y cable	1 per compressor channel	1 for the 76-KT (TRS to dual 1/4"; TRS to dual XLR if using its XLR jacks). Two dbx 1066s would need 4.
Monitor line cables	2	CONTROL ROOM out L/R → Peavey 260 inputs. 1/4" TRS to 1/4" TS.
Speaker cable	2 runs	Peavey 260 outputs → DIY monitors. Also needed between any guitar amp head and its cabinet. Never use an instrument cable for either.
Tascam 246 cables	2 + 2	MAIN MIX out L/R → 246 line inputs (2 x 1/4"). 246 line out L/R → ch 23-24 (2 x RCA-to-1/4" if its outs are RCA).
DJ mixer cables	2	DJ mixer master or booth out → PV6 inputs. Ends to match both mixers (often RCA-to-1/4").
Instrument cables	2 long, 1 medium, patch cables	Guitar → pedalboard → Diamond Vibrato → Phase 90 → amp; bass → bass amp.
Reference cable	1	3.5mm to dual RCA: phone or computer → 2-TK input.
Long balanced cables	2 (optional)	STUDIO OUTPUT L/R → garage PA. Length to suit the run; ends to match the PA's inputs.

Studio Connection Diagram: Signal Flow

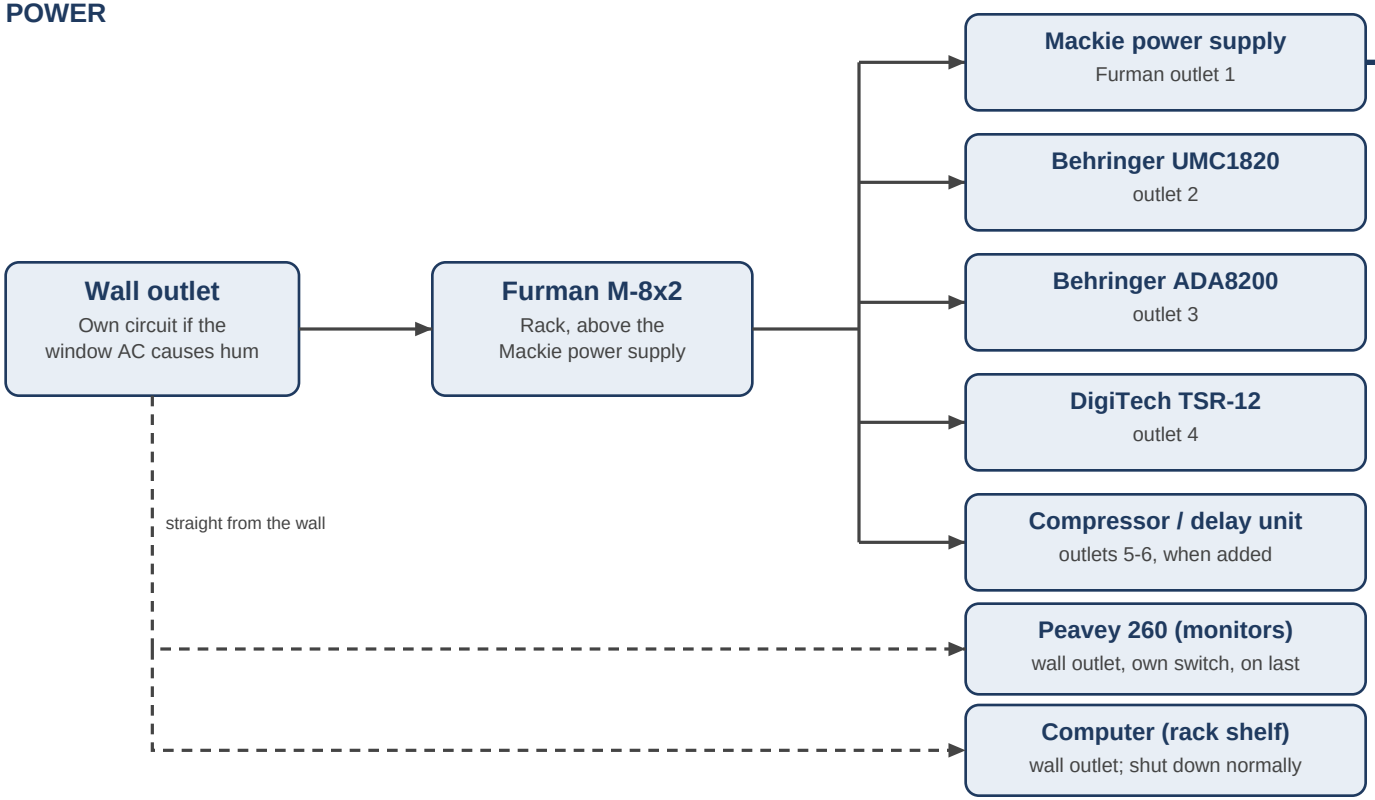
Mackie 24•8 studio, October 2026. Follow the arrows from sources to the console, rack and Reaper.



Studio Connection Diagram: Power, Instruments and Snake Colors

Everything plugs into the Furman. Instruments run through pedals before the amp mic.

POWER



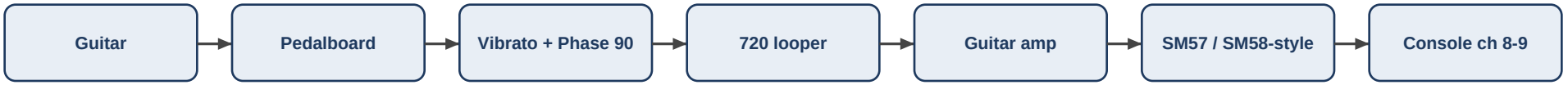
Keep the umbilical away from the snakes.

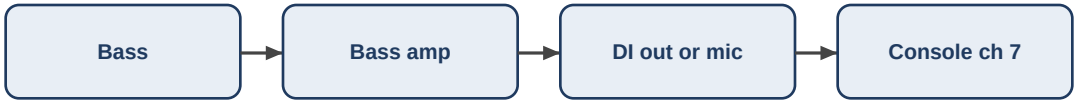
Power-up order

1. All levels down
2. Furman (rack comes up together)
3. Computer, Reaper
4. Wait 20-30 seconds
5. Peavey 260, own switch
6. Levels up

Power down in reverse order.
Phones and control room levels down before switching anything.

INSTRUMENT CHAINS





SH-201 L out > pedalboard > ch 17 (synth volume low).
Marshall: time-based pedals can go in its FX loop instead.
Amp head + separate cab? Speaker cable only.

SNAKE COLOR ORDER (both ends, all four snakes)

1	Brown	2	Red	3	Yellow	4	Green	5	Blue	6	Gray	7	White	8	Black

Tape near each end: white = recording, black = playback 1-8, blue = direct outs, yellow = ADA8200 playback 9-16.

Monitoring: Every Output and Where It Goes

The 24•8's monitor section explained button by button, every output the console has and what feeds it, and ready-made setups for the sessions you'll actually run. Button names are from the Mackie 8•Bus manual.

Every output the console has

Output (rear panel)	What's on it	Where it goes in this studio	Level control
CNTRL RM OUT L/R	Whatever source buttons are pressed in the MONITOR section, summed if MONO is in	Peavey 260 → DIY monitors (the room speakers)	CNTRL RM knob. Talkback drops it 20 dB.
STUDIO OUT L/R	The same monitor-section selection as the control room	Garage PA (powered mixer) when wanted	STUDIO knob. Talkback does NOT drop it.
PHONES 1	Its own five source buttons (below)	Your headphones (MDR-7506)	PHONES 1 level
PHONES 2	Its own five source buttons	Performer headphones (HD 280 Pro)	PHONES 2 level
MAIN MIX OUT L/R	The L/R mix: everything assigned L/R, at the main fader	Tascam 246 inputs (for tape). Not used for speakers.	Main L/R fader
MIX-B OUT L/R	The MIX-B bus: Reaper playback from the tape returns	Spare. Could feed a playback-only cue later.	MIX-B LEVEL
AUX SEND 1-6	Per-channel aux knobs, at each aux master	1: TSR-12. 2: open. 3/4 and 5/6: performer mixes (also selectable on the phones)	Aux masters
GROUP OUTS 1-8, DIRECT OUTS	Recording feeds	UMC1820 and ADA8200 (see the Setup Map)	Group masters; channel faders

The MONITOR section, button by button

- **Source buttons: L/R MIX, MIX-B, 2-TK (tape), EXTERNAL.** They pick what the control room and studio outputs carry, and you can press any combination; the sources add together. L/R MIX = the live mix at the main fader. MIX-B = Reaper playback coming back on the tape returns. 2-TK = the 2-TRACK INPUT jacks (your reference cable from a phone or computer). EXTERNAL = the EXTERNAL INPUT jacks (a spare stereo line in).
- **MONO.** Sums left and right for the speakers and the L/R meters. Press it to check that nothing disappears in mono, or to make hard-panned group channels sit in the middle of your headphones while tracking.
- **CNTRL RM level.** The room speaker volume, through the Peavey. Fully clockwise is unity; normal listening is well below that. Has no effect on headphones or on the recording.
- **STUDIO level.** Volume of the STUDIO OUT, which carries the same selection as the control room. In this studio that's the garage PA feed; leave it down when the garage isn't in use.
- **MIX-B section: MIX-B LEVEL and ASSIGN TO L/R MIX.** MIX-B LEVEL sets how loud Reaper playback is on the MIX-B bus. ASSIGN TO L/R MIX folds the playback into the main mix, which you'd use to hear a Reaper track while recording to tape, and should otherwise leave out so playback doesn't end up in a tape bounce by accident.

The PHONES sections (Phones 1 and Phones 2)

- Each has its own level knob, its own SOLO button, and five source buttons: **MONITOR, MIX-B, AUX SEND 3&4, AUX SEND 5&6, EXTERNAL.** You can press more than one.
- **MONITOR** = the same thing the control room speakers are getting, taken after the source buttons. The CNTRL RM knob and MONO do not touch the phones, so you can turn the room down to nothing and still hear everything in headphones.
- **MIX-B** = Reaper playback only. **AUX SEND 3&4** and **5&6** = the performer mixes built on those aux knobs (aux 3 left ear, aux 4 right). **EXTERNAL** = the EXTERNAL INPUT jacks, a private cue that appears on no channel.
- **Solo and headphones.** The solo bus only reaches a phones output whose MONITOR button is pressed. If Phones 2 is on AUX SEND 3&4, pressing SOLO lights the Rude Solo light but the performer hears no change, which is exactly what you want: you can solo-check a channel without disturbing them. Solo is after the fader on this console; the SOLO level knob at its center detent makes a soloed channel match its unsoloed level.

Talkback

- The talkback mic is built into the console, just above the main L/R fader. Four momentary buttons send it to **AUX SEND 1, AUX SEND 2, TAPE SUBGRPS** (the L/R mix and all eight groups) and **PHONES/STUDIO**.
- To talk to a performer on Phones 2: hold PHONES/STUDIO. To talk to the garage: the same button, since it also feeds the STUDIO OUT.
- While talkback is held, the control room output drops 20 dB so the speakers don't feed back into the mic. The studio output is not dropped. Keep the garage PA modest if someone's talking back from out there.
- Never hold AUX SEND 1 or 2 or TAPE SUBGRPS during a take: AUX 1 would put your voice into the reverb, and TAPE SUBGRPS would put it into the recording.
- The talkback level knob starts at its center detent.

Setups for common sessions

Session	MONITOR section	Phones 1 (you)	Phones 2 (performer)	Notes
Tracking a performer, you at the desk	L/R MIX in. MIX-B in if they're playing to a Reaper track. MONO out.	MONITOR	AUX SEND 3&4, built from aux 3/4 (PRE down) on the channels they need, plus MIX-B if they need the playback	Room speakers low or off while mics are open, so they don't bleed into the mics. Talk with PHONES/STUDIO.
Tracking yourself	L/R MIX in, MIX-B in	MONITOR, and MONO pressed in the monitor section if the hard-panned groups sound lopsided	Unused or same as Phones 1	MONO changes only what you hear, never the recording.
Full band, everyone in headphones	L/R MIX and MIX-B in	MONITOR	AUX SEND 3&4 for the second person; everyone else on a personal mixer fed from aux 3-6	Each aux pair is one stereo mix. The personal mixers are the headphone amps for players 3 and up.
Mixing in Reaper	MIX-B in only. Everything else out.	MONITOR, or nothing: use the speakers	Off	You're listening to Reaper's output on the tape returns. The console's channel faders are out of the path. Check in mono with MONO.
Reference track A/B	Toggle between MIX-B (your mix) and 2-TK (the reference). Press only one at a time.	MONITOR	Off	Match the two levels by ear with MIX-B LEVEL and the phone's volume before judging anything.
Recording to the Tascam 246	L/R MIX in. Keep ch 23-24 out of L/R.	MONITOR	Off	The 246 gets MAIN MIX OUT at the main fader. To hear the tape back afterwards, assign ch 23-24 to L/R and listen on L/R MIX; MIX-B has nothing to do with the tape.
Playback in the garage	Same selection as the room; STUDIO knob up	MONITOR	Off	STUDIO OUT mirrors the control room selection. Talkback reaches it undimmed.

Rules that keep you out of trouble

- Only the MONITOR section and the two PHONES sections decide what people hear. Nothing in them changes what Reaper records, so experiment freely.
- Open mics plus loud room speakers equals bleed and feedback. While tracking, monitor on headphones and keep CNTRL RM low; the Peavey's own knobs are the last line of defense.
- Press one of L/R MIX or MIX-B at a time when you need to know which is which. With both in, a Reaper track and the live mix add together, which is right for overdubs but confusing for troubleshooting.
- If you hear nothing in the room: check in this order: CNTRL RM knob, a source button pressed, Peavey level knobs, Peavey power. If you hear nothing in Phones 2: its level, a source button, and whether the aux 3/4 masters are up.
- If the performer hears your solo-checking, Phones 2 is on MONITOR. Switch it to AUX SEND 3&4.
- If talkback feeds back: you're holding it with CNTRL RM too loud for a 20 dB dip, or the garage PA is loud enough to reach the mic. Turn the room down first.

Power: Furman Outlets, Start-Up and Shutdown

Where everything plugs in, the order to switch it on and off, and why. The Mackie manual's only power rule is that the 24•8 is powered only by its own Mackie 8•Bus power supply; the sequence below is standard studio practice, not a Mackie requirement.

Where the Furman plugs in

- **Straight into a grounded wall outlet.** Not into a power strip, surge protector or UPS. Stacking surge protectors is against most electrical codes and UL listings, can make the surge protection work worse rather than better, and puts the whole rack's load through the weakest strip's wiring and switch.
- Check the outlet first with a plug-in outlet tester (a few dollars at a hardware store): it should show correct wiring and a good ground. The Furman's surge protection and filtering both dump to ground; on an ungrounded or miswired outlet they do little, and that's also a common cause of hum.
- If the wall outlet is too far from the rack, use a **heavy-duty extension cord**, 12 or 14 gauge, three-prong, rated 15 A, as short as will reach. That's acceptable where a power strip isn't. Avoid thin household cords.
- Use an outlet on a different circuit from the window AC if you can (find out by flipping breakers and seeing what goes dark). The AC's compressor kicking on is the most likely source of clicks in this room.
- The Peavey 260 and the computer, which stay off the Furman, go to the wall too, on the same circuit as the Furman so everything shares one ground. Plugging those two into an ordinary surge strip is fine, since that strip isn't stacked on anything.

What the Furman M-8x2 gives you

- Nine outlets: eight switched outlets on the rear panel and one convenience outlet on the front. Fifteen-amp rating with its own circuit breaker, surge and spike protection, and RFI/EMI noise filtering. A Protection OK light on the front tells you the surge protection is still alive; if it goes dark, the Furman has taken a hit and should be replaced.
- The front switch turns the eight rear outlets on and off together. Everything plugged into the back comes up at once, so the Furman is step one of the start-up and the last thing off.
- Load check: the whole rack draws well under 15 A. The Mackie power supply is the biggest item at around 2 A; the UMC1820, ADA8200, TSR-12 and a compressor are a fraction of an amp each. The Peavey 260 power amp is the one thing to keep off the Furman's switched outlets (see below).

Which outlet gets what

Rear outlet	Plug in	Why here
1	Mackie 24•8 power supply	Biggest draw, and the console must be up before anything that listens to it.
2	Behringer UMC1820	Computer interface. Fine on a switched outlet; the computer wakes it over USB.
3	Behringer ADA8200	Comes up with the UMC1820 so the ADAT clock locks right away.
4	DigiTech TSR-12	Outboard effects.
5	Compressor (76-KT) when it arrives	Outboard dynamics.
6	Delay / second effects unit, when added	Outboard effects.
7	Spare	Patchbays need no power. Keep this free.
8	Spare	Keep one free. Never daisy-chain a second power strip off the Furman.
Front	Phone or laptop charger, soldering iron, a work light	Convenience only. Nothing in the audio path.
Not on the Furman	Peavey 260 power amp; the computer	See the next section.

Label each plug at the Furman end with masking tape ("MACKIE PSU", "UMC", and so on) so a tripped breaker or a swap doesn't turn into a guessing game.

Two things that stay off the Furman's switched outlets

- **Peavey 260 power amp.** A power amp must come on last and go off first, so it can't be on the same switch as everything else: if it powered up with the rack, the turn-on thumps of the console and interface would go straight to the speakers. Plug it into a wall outlet on the same circuit (or a separate surge strip), and use its own power switch as the final step each way.
- **The computer.** It sits on a shelf in the rack but does not plug into the Furman: cutting its power with the Furman switch is the same as pulling the plug mid-session. Give it its own outlet and shut it down normally. If the Furman trips, the computer stays up and Reaper's autosave is intact.
- Everything else audio belongs on the Furman: the point of a conditioner is that one filtered, surge-protected feed covers the whole signal chain, which also helps against ground-loop hum.

Start-up, in order

Step	Switch on	Before you do
1	All levels down: CNTRL RM, STUDIO, both PHONES, Peavey 260 input knobs. Main L/R fader down.	Nothing can thump through speakers or headphones that have no level.
2	Furman front switch. The eight rear outlets come up: Mackie power supply, UMC1820, ADA8200, TSR-12, compressor, delay unit.	Check the Protection OK light. The console's power LED and the power supply fan should come on.
3	Computer, then open Reaper.	The UMC1820 should show up as the audio device; the ADA8200 SYNC light should read LOCKED.
4	Wait about 20-30 seconds.	Lets the console's circuits settle and any relays click before the amp is live.
5	Peavey 260 power switch (its own switch, at the wall).	Its level knobs are still down from step 1.
6	Bring up CNTRL RM to a normal setting, then the Peavey's level knobs, then phones levels.	Now the room and headphones are live. Raise the main L/R fader last, with music playing, so you hear it come in.

Shutdown, in reverse

Step	Switch off	Why
1	Save the Reaper project. Main L/R fader down, CNTRL RM, STUDIO and PHONES levels down, Peavey level knobs down.	Protects the session and the speakers from whatever happens next.
2	Peavey 260 power switch off.	Amp off first, so no later click reaches the speakers.
3	Close Reaper and shut the computer down from its menu.	The interface will drop off the USB cleanly.
4	Furman front switch off. The console, interface, ADA8200 and outboard all go down together.	Fine for all of this gear; none of it has a disk or a session to lose.
5	Leave the Furman plugged into the wall.	Its surge protection only works while it's connected. Unplug it only for storms or when you're away for a long time.

Details that matter

- The 24•8 has no power switch of its own; it's on when the power supply is on. The power supply's umbilical plugs into DC POWER IN on the console: connect it with everything off, and never plug or unplug it live.
- Mackie's one hard rule: power the console only from the Mackie 8•Bus power supply. No substitutes.
- Phantom power: switch the 48V blocks on after start-up with the block's faders down, and off before shutdown. Never plug or unplug a mic on a powered block.
- The power supply runs warm and its fan needs air: keep the empty rack space between it and the Furman, and never block the front vents (the kitten's spot is fine; a blanket is not).
- If the Furman's breaker trips: everything on the rear outlets goes dark at once, but the Peavey and computer stay up. Turn the Peavey's levels down, press the breaker back in, and start again from step 2. If it trips twice, something is drawing too much or has a fault; stop and find it.
- Storms: the Furman absorbs surges, but nothing stops a direct lightning hit. Unplug the Furman and the Peavey from the wall when a bad storm is coming.
- If you ever add a power sequencer (Furman's M-8S model does this), it automates steps 2-5 with one button: conditioner, then outboard, then amps, a few seconds apart, and in reverse on shutdown. Not needed now.

Gain Staging Guide

Channel-by-channel levels for the Mackie 24•8 studio, October 2026. Goal: clean, strong signals with room to spare, peaking around -12 to -6 dBFS in Reaper.

1. Zero everything first

- Trims all the way down, aux sends down, EQ switched out, phones and control room levels down.
- Group masters 1-8 and the main L/R fader at U (unity).
- UMC1820 inputs 1-8: LINE, not INST; gain knobs at minimum; pads off (switch a pad in only if an input still clips). ADA8200: gain knobs at minimum. The console does the amplifying; these just pass it along.
- Reaper track faders at 0 dB.

2. Set each channel (one at a time)

- Put the channel's fader at U and make sure it isn't muted. Solo on this console is taken after the fader, so the fader has to be at U for the reading to mean anything.
- Have the player perform the loudest part of the song, not a quiet warm-up.
- Press the channel's SOLO and watch the L/R meters. A hard-panned channel shows on one meter only.
- Raise the trim until peaks regularly reach 0 on the meters. An occasional +2 or +3 is fine; steady red is not.
- Release SOLO. Leave the fader at U; on group channels, the group master stays at U too.
- Check Reaper: peaks around -12 to -6 dBFS. Too hot or too quiet? Change the trim, not the fader.

3. Channel chart

Starting trims are where to begin, not final settings; your ears and meters decide. U = unity (0) on the fader.

Ch	Source	Input	48V	Starting trim	Fader	Notes
1	Kick (kick mic)	Mic	If cond.	Low	U	Very loud source. Group 1.
2	Snare (snare mic)	Mic	If cond.	Low	U	Check rimshots; they peak hardest. Group 2.
3	Rack tom (SM58-style)	Mic	No	Low-med	U	Hit hard to set level. Groups 3-4.
4	Floor tom (SM58-style)	Mic	No	Low-med	U	Groups 3-4.
5	Room L	Mic	If cond.	Med-high	U	Set L and R to the same trim. Groups 3-4.
6	Room R	Mic	If cond.	Med-high	U	Match ch 5. Groups 3-4.
7	Bass amp DI out (or mic)	Mic or line	No	Low-med	U	If the DI out is line level, use the line input. Group 5.
8	Magnatone (SM57)	Mic	No	Low-med	U	Depends on amp volume; set the amp first. Group 6.
9	Marshall MG250DFX or Orange (SM58-style)	Mic	No	Low-med	U	Shares group 6 with ch 8. Set each trim first, then balance the two on their faders and leave them.
10	Lead vocal (Worker Bee)	Mic	Yes	Med	U	Set with the loudest line of the song. Direct out → ADA8200 in 1.
11	Wireless (ATW-R3210)	Mic	No	Med	U	Set the receiver's output level first. Direct out → ADA8200 in 2.
12-13	Reface CS L/R	Line	No	Low	U	Synth volume about 3/4. Groups 7-8.
14	Trumpet (SM58)	Mic	No	Low	U	Very loud; player 1-2 ft back. Direct out → ADA8200 in 3.
15	Vocal 2 (Peavey)	Mic	No	Med-high	U	Direct out → ADA8200 in 4.
16	Vocal 3 (Turner 95D)	Mic	No (see note)	High	U	Low-output vintage mic; check impedance first. Direct out → ADA8200 in 5.
17-18	Roland SH-201 L/R	Line	No	Low	U	Through the pedalboard, check levels again. Direct outs → ADA8200 in 6-7.

Ch	Source	Input	48V	Starting trim	Fader	Notes
19-20	PV6 main out (e-drums, SamplePad 4, DJ)	Line	No	Low	To taste	PV6 at unity. Monitor only.
21	1970s Roland drum machine	Line	No	Low-med	U	Vintage outputs can be quieter. Direct out → ADA8200 in 8.
22	Open	—	—	—	—	
23-24	Tascam 246 line out L/R	Line	No	Med	To taste	Consumer-level outputs; trim up a little more than other line sources.

48V (phantom power) is switched in blocks of eight channels: 1-8, 9-16 and 17-24, with the switch at the top of each block. "If cond." means only if that mic is a condenser. The Worker Bee on ch 10 needs it, so the mic inputs on 9-16 all get 48V. That's safe for balanced dynamic mics (SM58, Peavey). Check the wireless receiver's manual before leaving it on a powered block. Keep the Turner 95D off a powered block until it's on a matching transformer or proper low-impedance XLR wiring. Turn the block's faders down before switching phantom, and don't plug or unplug mics with it on.

4. What ends up in the recording

- Group channels: trim, EQ, fader, pan and the group master all shape what Reaper gets.
- Direct-out channels: trim, EQ, fader and mute. Pan and the L/R button don't matter to the recording.
- So set EQ and faders before the take and leave them. Use the phones level, MIX-B and aux sends to change what people hear.

5. Outboard and other gear

- **Compressor (76-KT or dbx):** see the Setting Compression Levels page. Short version: trim first with the compressor bypassed, then Input (or Threshold) for 3-6 dB of gain reduction, then Output until bypass and un-bypass sound equally loud.
- **TSR-12:** aux 1 master at about noon; TSR-12 input so its meter peaks just under overload; mix 100% wet; AUX RETURN 1 to taste.
- **PV6 submixer:** its channel gains and master at unity; set each source's own volume moderately high; balance the sources on the PV6.
- **Tascam 246:** input trims low to start, then tape meters around 0 VU; a little past for saturation.
- **Reaper playback:** MIX-B level and tape-return levels set so playback sits naturally under the live mix. UMC1820 MAIN OUT knob all the way up, since it sets the level of playback tracks 1-2.
- **Headphones:** phones level low before anyone puts headphones on; build mixes with aux sends 3-6.
- **Garage PA:** STUDIO level low, PA inputs at unity, then raise STUDIO level.
- **Peavey 260 + DIY monitors:** CONTROL ROOM level and Peavey level knobs low; play a mix, raise CONTROL ROOM to a normal setting, then raise the Peavey until it's loud enough. VU meters should stay below the red.

6. Final checks

- Re-check with the whole band playing: everyone plays louder together.
- Hiss on a quiet singer: move the mic closer before adding gain. If the trim is near max and hiss remains, consider an inline booster (Cloudlifter-style) for that mic.
- Distortion with meters fine: check the source itself (amp, synth, DJ mixer, PV6) for clipping.
- Clicks and pops only on Reaper inputs 9-16: check the ADA8200's clock setting (see the Setup Map).
- Write down final trim positions (masking tape on the console) so sessions are repeatable.

Setting Compression Levels

How to dial in the Klark Teknik 76-KT on a 24•8 channel insert, plus settings for the dbx alternatives. The goal: the compressor does what you want, and the recording stays at the same level as before.

What the knobs do (76-KT)

- **Input:** how hard the signal hits the compressor. On an 1176-style unit there is no threshold knob; turning Input up is what makes it compress more.
- **Output:** brings the level back up after compression. This is the knob that keeps the recording at the same volume as bypass.
- **Attack:** how fast it grabs a loud note, about 20 to 800 microseconds. Clockwise is faster. Fast = grabs the drum hit itself; slower = lets the hit through and clamps what follows, which keeps punch.
- **Release:** how long it holds on after the note drops, about 80 to 700 milliseconds. Clockwise is faster. Fast release sounds pumping and aggressive; slow sounds smoother.
- **Ratio buttons:** 4:1 moderate, 8:1 severe, 12:1 mild limiting, 20:1 hard limiting. Pressing all four at once ("all buttons in") gives the crushed, exploding sound on drums and room mics.
- **Meter:** shows gain reduction (GR), the amount in dB being turned down. Set the meter switch to GR for all of the steps below. The +4/+8 positions show output level instead.

Wiring and gain structure

- Channel insert jack → insert Y cable: TRS end into the insert, tip (send) to the 76-KT input, ring (return) from the 76-KT output. The compressor sits after the channel trim and before the EQ and fader, so it's recorded.
- The 76-KT's XLR jacks are balanced, +4 dBu. The Mackie inserts are unbalanced, -2 dBu, a little lower. That's fine: use a TRS-to-dual-XLR insert cable, or the 76-KT's 1/4" jacks if your unit has them. Expect to run the 76-KT Input a bit higher than you would from a +4 source.
- Set the channel trim first, exactly as in the Gain Staging Guide, with the compressor bypassed. The compressor must never be used to fix a bad trim setting.

Step by step

Step	What to do	What you're listening or looking for
1	Bypass the 76-KT (or pull the insert). Play the part at full volume and note where the channel meters and Reaper peak.	This is the level you'll match at the end.
2	Un-bypass. Output at about 12 o'clock, Input low, Attack and Release at 12 o'clock, ratio 4:1, meter on GR.	The meter needle should sit at 0 with the part playing: no compression yet.
3	Turn Input up slowly while the part plays.	The needle starts dipping left on loud notes. Stop when it dips to the amount in the table below.
4	Set Attack. Start at 12 o'clock, turn it slower (counter-clockwise) if the drum or pluck loses its snap.	Transients still cut through; the needle moves right after the hit, not on it.
5	Set Release. Turn it faster (clockwise) until the needle gets back to 0 between notes, then back off a touch.	The needle breathes with the music and never gets stuck down. Fast on drums, slower on vocals and bass.
6	Flip bypass in and out while adjusting Output.	Both sound equally loud. The console meters and Reaper peak where they did in step 1.
7	Check the whole band playing together, and check Reaper for clipping.	If the recording is now too hot, lower Output, not the trim.

Starting points

Source	Ratio	Attack	Release	Gain reduction	Notes
Lead vocal	4:1	Medium (12 o'clock)	Medium	3-6 dB on loud lines	Fast enough to catch peaks, slow enough that quiet words don't jump up. 8:1 for a shouted take.
Bass (DI or amp)	4:1	Slow (9-10 o'clock)	Medium-slow	3-5 dB	Slow attack keeps the pluck; the compressor evens out note-to-note volume.

Source	Ratio	Attack	Release	Gain reduction	Notes
Room mic (Turner or ch 5/6)	All buttons in	Fast (3 o'clock)	Fast (3-4 o'clock)	10-20 dB, pinned	The Tame Impala crush. Blend this under the close drum mics in Reaper.
Kick or snare (close mic)	4:1 or 8:1	Slow (9 o'clock)	Fast	3-6 dB	Slow attack lets the hit through; fast release pumps. Too fast an attack makes drums sound dull.
Drum pair (groups 3-4, if a 2-channel unit)	4:1	Medium-slow	Fast	4-8 dB	Link the two channels so the stereo image doesn't wander.
Acoustic guitar	4:1	Medium	Medium	2-4 dB	Light touch; too much flattens the strumming.
Synths through pedals	4:1 or 8:1	Fast	Medium	3-8 dB	Tames the spiky peaks from fuzz pedals before the trim.

Knob positions are rough: the 76-KT's attack and release scales run backwards from most compressors (clockwise = faster), so go by the meter and your ears.

If you buy the dbx alternative instead (266XS or 1066)

• These have a Threshold knob, so Input stays fixed: leave the console insert level alone and turn Threshold down until the gain-reduction meter shows the amount in the table above.
• Ratio knob: 4:1 for vocals, bass and acoustic; 8:1 or higher for drums. OverEasy (soft knee) on for vocals and bass; off (hard knee) for drums.
• Attack and Release: the Auto mode is a fine default for vocals and bass; switch to manual for drums and set them as in the table.
• Output gain: match bypass exactly as in step 6. The gate section: threshold all the way down (off) unless you're using it to cut tom ring; then raise it just until the ring stops between hits.
• Stereo link for the drum pair: press Stereo Couple and use channel 1's controls for both.

Common mistakes

• Needle pinned left all the time: Input is too high or release too slow. The part will sound squashed and small. Back Input off until the needle returns to 0 between phrases (unless you're going for the room-mic crush on purpose).
• Drums sound dull: attack is too fast. Slow it down until the stick hit comes back.
• Everything got louder when you un-bypassed: Output is too high. That's not a better sound, it's just louder. Match bypass before judging.
• Hiss or noise rises between notes: too much gain reduction with a fast release is pumping the noise floor up. Reduce Input, or add the gate (DN514+) on that channel.
• Mix bus: don't put the 76-KT on the main L/R inserts. Reaper's ReaComp or a bus-compressor plugin does that job at mixdown, where you can undo it.
• Hardware compression is printed to the recording. When unsure, use less; you can always add more in Reaper, but you can't take it off.

EQ Settings, Channel by Channel

Starting points for the 24•8's channel EQ, drawn the way the knobs look on the board. Orange = moved from flat. These are places to begin while tracking, not rules; your mics, room and ears decide.

The EQ section on each 24•8 channel, top to bottom

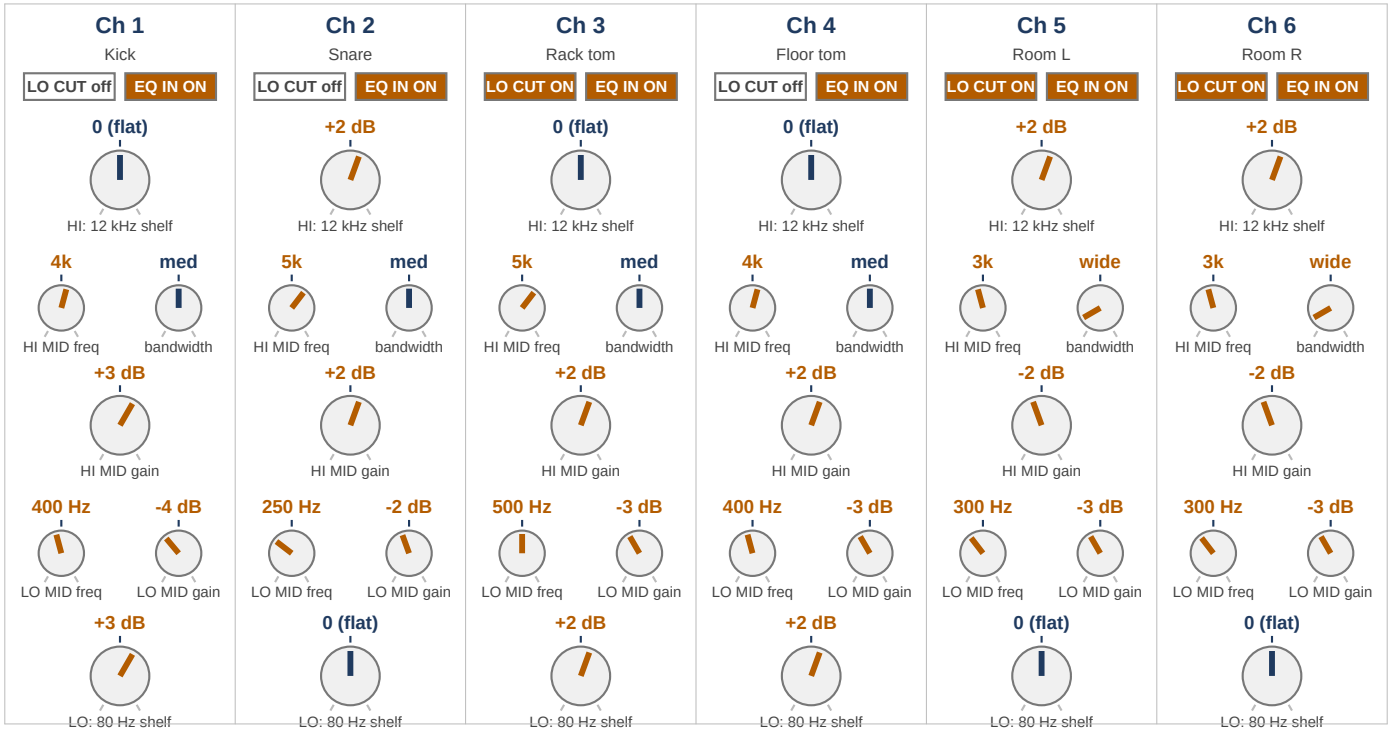
- **LO CUT button:** a high-pass filter at 75 Hz, 18 dB per octave. Removes rumble, floor thumps, plosive pops and mic-stand noise below 75 Hz. It works even when EQ IN is off. **Use it on** vocals, toms, guitar amps, trumpet and room mics. **Never on** kick, bass, synths with low notes, or the drum machine: it takes the bottom off them.
- **HI knob:** a 12 kHz shelf, ±15 dB. Everything above about 12 kHz goes up or down together. **For:** air and sparkle on vocals, cymbals and acoustic guitar. **Don't:** boost it on dynamic mics that are already hissy, or on a trumpet; a 15 dB range is there for repair, 1-3 dB is a normal move.
- **HI MID knobs:** three of them. FREQ sweeps the center from 500 Hz to 18 kHz; BANDWIDTH sets how wide the band is, from 3 octaves (wide, gentle) to 1/12 octave (a surgical notch); the gain knob is ±15 dB. This is the presence band: 2-5 kHz is where a vocal or guitar cuts through, and also where things get harsh. **Don't:** boost a narrow band; narrow settings are for cutting problems (a ring, a whistle). Boosts should be wide.
- **LO MID knobs:** FREQ sweeps 45 Hz to 3 kHz and the gain is ±15 dB; the width is fixed at 2 octaves (broad). This is the mud band: 200-500 Hz is where drums get boxy and vocals get chesty. Most useful EQ moves on this console are small cuts here. **Don't:** boost it on more than one or two sources, or the mix turns to soup.
- **LO knob:** an 80 Hz shelf, ±15 dB. Weight and thump. **For:** kick, floor tom, bass. **Don't:** boost it on anything that isn't a bass instrument; low end piles up fast, and the cheap fix is LO CUT on everything that doesn't need lows.
- **EQ IN button:** switches the four EQ bands in or out (LO CUT stays active). Leave it out on any channel set flat, and use it to A/B your EQ: if "in" doesn't sound clearly better, set the knobs back to 0.

Rules that matter more than any setting

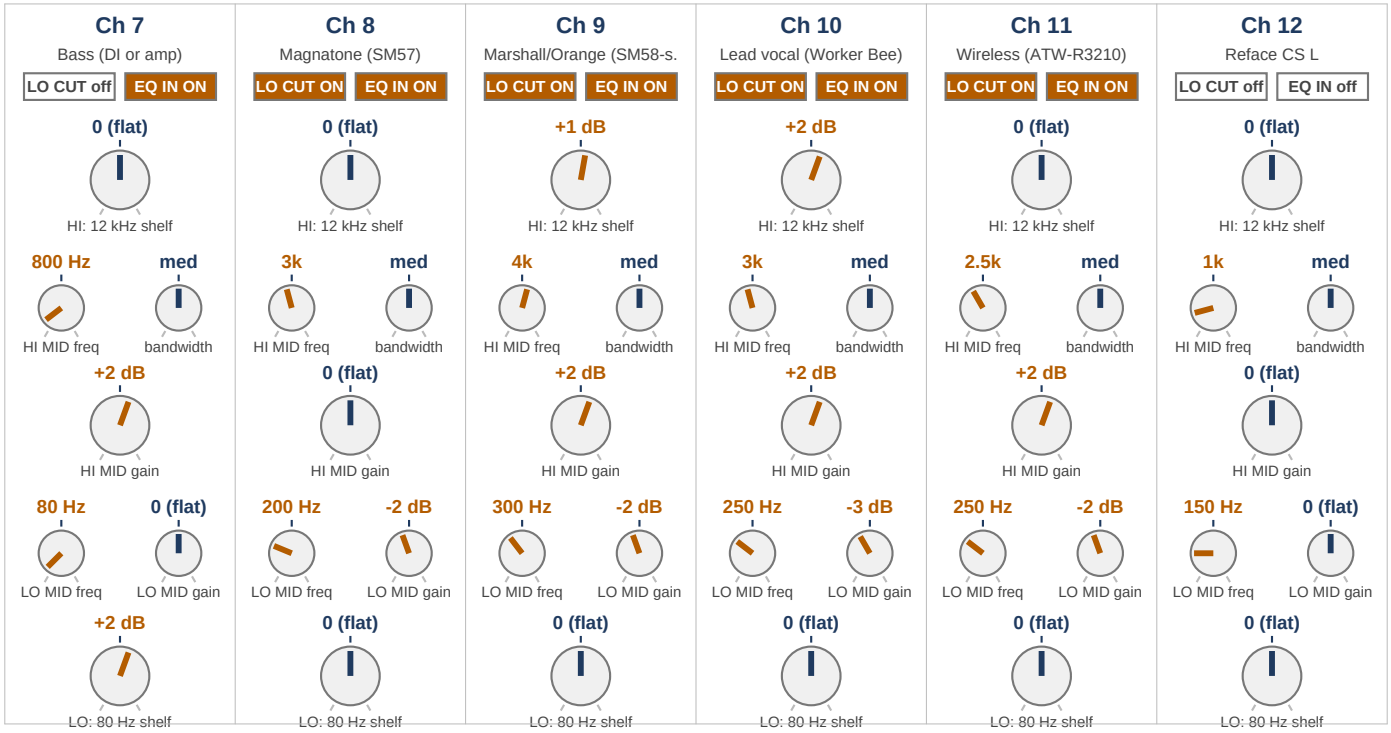
- Set trim and the compressor first, then EQ. The EQ is before the fader and the group outs and direct outs, so whatever you do here is recorded. Reaper's ReaEQ can do the fine work later with undo; the console EQ is for broad shaping and fixing real problems.
- Cut before you boost. A 3 dB cut at 300 Hz does what a 3 dB boost at 3 kHz does, with less noise and less chance of clipping.
- Find a problem by sweeping: boost the mid gain, turn the FREQ knob slowly until the ugly part jumps out, then turn the gain down to a cut at that spot.
- EQ moves of 1-3 dB are normal. If you need more than 6 dB, move the mic instead: an inch at the amp grille or a few inches on a vocal changes more than any knob.
- Which mic is on the channel matters. A condenser (Worker Bee) has more top and more low rumble: LO CUT on, go easy on HI. Dynamic mics (SM57, SM58, the knockoffs, Peavey) are a bit darker: small presence lifts are fine. The Turner 95D has almost no top or bottom to begin with. Line sources (synths, PV6, Tascam) usually want no EQ at all.
- Stereo pairs (rooms 5-6, Reface 12-13, SH-201 17-18, Tascam 23-24) get identical settings, or the image shifts.
- Listen in context. Solo makes everything sound like it needs more of everything. Decide with the band playing.

Channels 1-6

Each strip reads top to bottom like the board: LO CUT and EQ IN buttons, HI shelf, HI MID (frequency, bandwidth, gain), LO MID (frequency, gain), LO shelf. Orange = set away from flat. Pointer straight up = 0 dB or the middle of the sweep.



Channels 7-12



Channels 13-18

Ch 13	Ch 14	Ch 15	Ch 16	Ch 17	Ch 18
Reface CS R	Trumpet (SM58)	Vocal 2 (Peavey)	Vocal 3 (Turner 95D)	SH-201 L	SH-201 R
LO CUT off	LO CUT ON	LO CUT ON	LO CUT ON	LO CUT off	LO CUT off
EQ IN off	EQ IN ON	EQ IN ON	EQ IN ON	EQ IN off	EQ IN off
0 (flat)	-1 dB	+1 dB	+3 dB	0 (flat)	0 (flat)
HI: 12 kHz shelf	HI: 12 kHz shelf	HI: 12 kHz shelf	HI: 12 kHz shelf	HI: 12 kHz shelf	HI: 12 kHz shelf
1k	3k	3k	4k	1k	1k
HI MID freq	HI MID freq	HI MID freq	HI MID freq	HI MID freq	HI MID freq
med	narrow	med	med	med	med
bandwidth	bandwidth	bandwidth	bandwidth	bandwidth	bandwidth
0 (flat)	-3 dB	+2 dB	+3 dB	0 (flat)	0 (flat)
HI MID gain	HI MID gain	HI MID gain	HI MID gain	HI MID gain	HI MID gain
150 Hz	300 Hz	250 Hz	400 Hz	150 Hz	150 Hz
LO MID freq	LO MID freq	LO MID freq	LO MID freq	LO MID freq	LO MID freq
0 (flat)	0 (flat)	-2 dB	-3 dB	0 (flat)	0 (flat)
LO MID gain	LO MID gain	LO MID gain	LO MID gain	LO MID gain	LO MID gain
0 (flat)	0 (flat)	0 (flat)	0 (flat)	0 (flat)	0 (flat)
LO: 80 Hz shelf	LO: 80 Hz shelf	LO: 80 Hz shelf	LO: 80 Hz shelf	LO: 80 Hz shelf	LO: 80 Hz shelf

Channels 19-24

Ch 19	Ch 20	Ch 21	Ch 22	Ch 23	Ch 24
PV6 submix L	PV6 submix R	1970s drum machine	Open	Tascam 246 L	Tascam 246 R
LO CUT off	LO CUT off	LO CUT off	LO CUT off	LO CUT off	LO CUT off
EQ IN off	EQ IN off	EQ IN ON	EQ IN off	EQ IN ON	EQ IN ON
0 (flat)	0 (flat)	+2 dB	0 (flat)	+1 dB	+1 dB
HI: 12 kHz shelf	HI: 12 kHz shelf	HI: 12 kHz shelf	HI: 12 kHz shelf	HI: 12 kHz shelf	HI: 12 kHz shelf
1k	1k	3k	1k	3k	3k
HI MID freq	HI MID freq	HI MID freq	HI MID freq	HI MID freq	HI MID freq
med	med	med	med	med	med
bandwidth	bandwidth	bandwidth	bandwidth	bandwidth	bandwidth
0 (flat)	0 (flat)	+1 dB	0 (flat)	0 (flat)	0 (flat)
HI MID gain	HI MID gain	HI MID gain	HI MID gain	HI MID gain	HI MID gain
150 Hz	150 Hz	200 Hz	150 Hz	150 Hz	150 Hz
LO MID freq	LO MID freq	LO MID freq	LO MID freq	LO MID freq	LO MID freq
0 (flat)	0 (flat)	0 (flat)	0 (flat)	0 (flat)	0 (flat)
LO MID gain	LO MID gain	LO MID gain	LO MID gain	LO MID gain	LO MID gain
0 (flat)	0 (flat)	+2 dB	0 (flat)	0 (flat)	0 (flat)
LO: 80 Hz shelf	LO: 80 Hz shelf	LO: 80 Hz shelf	LO: 80 Hz shelf	LO: 80 Hz shelf	LO: 80 Hz shelf

Why each channel is set this way

Ch	Source	Why these settings
1	Kick	Low shelf adds thump; 400 Hz cut removes the cardboard "box"; 4 kHz gives the beater click so it cuts through. Keep LO CUT off.
2	Snare	A little top for crack. If it rings, sweep LO MID until the ring is loudest, then cut 3-4 dB there.
3	Rack tom	LO CUT on: nothing useful below 75 Hz on a 13" tom. Cut the boxy 500 Hz, add stick attack at 5 kHz.
4	Floor tom	Same idea as the rack tom. LO CUT off so the low thump stays. Both toms share groups 3-4, so keep them similar.
5	Room L	Rooms are about air and smack. Cut the mud at 300 Hz, soften 3 kHz if the cymbals are harsh, LO CUT on (the kick mic already has the lows).
6	Room R	Match channel 5 exactly, or the stereo picture shifts.
7	Bass (DI or amp)	Gentle: a touch of low shelf for weight, a little 800 Hz so it's audible on small speakers. No LO CUT ever on bass.
8	Magnatone (SM57)	SM57 on a tube amp needs almost nothing. LO CUT on to drop the rumble a close mic picks up. Cut a little 200 Hz if it's muddy.
9	Marshall/Orange (SM58-style)	SM58-style mics are a little darker than an SM57 on an amp: a small lift at 4 kHz and the HI shelf brings back the bite.
10	Lead vocal (Worker Bee)	Condenser: LO CUT on for plosives and rumble, cut the 250 Hz chestiness, a little 3 kHz presence, HI shelf for air. Cut less than you think.
11	Wireless (ATW-R3210)	Depends on what's plugged into the body pack. Start flat and add presence only if it sounds dull.
12	Reface CS L	Flat. Synths come out already shaped. EQ IN off until something is actually wrong.
13	Reface CS R	Match channel 12.
14	Trumpet (SM58)	Trumpet is naturally bright and loud. Cut 3 kHz a little so it doesn't pierce; LO CUT on. Distance from the mic does more than EQ.
15	Vocal 2 (Peavey)	Dynamic mic: a bit more presence than the Worker Bee needs, plus LO CUT.
16	Vocal 3 (Turner 95D)	Vintage mic with a narrow 100 Hz-10 kHz range. Boost the top to open it up, or leave it dark on purpose for the lo-fi vocal sound.
17	SH-201 L	Flat; EQ on the synth or the pedals instead. Through fuzz pedals, cut 3-5 kHz a little if it's harsh.
18	SH-201 R	Match channel 17.
19	PV6 submix L	Monitor only. Flat; adjust on the PV6.
20	PV6 submix R	Monitor only. Flat.
21	1970s drum machine	Old drum machines are dull and thin: a little low shelf and top brings them to life. Keep LO CUT off.
23	Tascam 246 L	Tape loses a little top; a 1 dB HI shelf brings it back. Otherwise flat, so you hear what the tape did.
24	Tascam 246 R	Match channel 23.

Mic Placement

Where each mic goes, how far, and aimed at what, for every mic'd source in the channel map. Distances are starting points; move the mic before you reach for EQ. Room dimensions aren't needed for most of this, since placement is about distance from the source. The places where your room sizes would change the advice are marked "room-dependent" and listed at the end.

Rules that cover everything

- **Move the mic, not the EQ.** An inch at a guitar grille or two inches on a vocal changes more than any knob on the console. Set placement with headphones on and someone playing, then set trim, then EQ.
- **Closer = more bass, less room.** Every directional mic you own (SM57, SM58, the knockoffs, Peavey, Worker Bee in cardioid) gets boomier as it gets closer, called proximity effect. Use it on purpose for warmth; back off when a source gets muddy.
- **Aim the mic's back at what you don't want.** Cardioid mics reject sound from directly behind. Point the back of the snare mic at the hi-hat, the back of the vocal mic at the drum kit, and bleed drops without moving anything else.
- **3:1 rule for two mics on one source.** When two mics can hear the same thing (two tom mics, the room pair, a vocal mic near the acoustic guitar mic), keep them at least three times as far from each other as each is from its source, or their sound combines with hollow phase cancellation.
- **Check in mono.** Press MONO in the console's monitor section with the room pair up. If the drums get thinner or the kick disappears, a mic is out of phase: move it a few inches or flip a cable's polarity.
- **Stands and cables.** Boom arm over the heavy tripod leg so it doesn't tip. One loop of cable around the stand. Never let a mic cable hang from the mic's connector.

Drums

Mic / channel	Where	Distance	Aim	Notes
Kick (ch 1)	Inside the kick through the port hole if the front head has one; otherwise 2-4 in. outside the front head, off-center	Inside: 3-6 in. from the beater head. Outside: 2-4 in.	At the beater spot for click; angled toward the shell for more boom	More click = closer to the beater. Take the pillow out of the drum before you blame the mic. Room-dependent: no.
Snare (ch 2)	Top head, over the rim, between the hi-hat and the first rack tom	1-2 in. above the rim, pointed at the center	Capsule aimed at the center of the head, body tilted so its back faces the hi-hat	If the snare sounds thin, lower the angle; if too much hi-hat, rotate the back of the mic toward it. Stand on the drummer's left.
Rack tom (ch 3)	Over the top rim, away from the cymbal above it	1-2 in. above the head, at the rim	At the center of the head	SM58-style: pull the grille basket off for a closer, punchier position if it's a true 58 clone.
Floor tom (ch 4)	Over the top rim on the side away from the ride	1-2 in. above the head	At the center	Same as the rack tom. Check the 3:1 rule between the two tom mics.
Room L / R (ch 5-6)	Room-dependent. Default: a spaced pair, 6-10 ft in front of the kit, 4-6 ft apart, at 5-6 ft high, equidistant from the snare	6-10 ft from the kit	Both at the snare	Measure from the snare to each mic with a tape and make them equal: that keeps the snare centered. The further back, the more room sound; in a small room, 6 ft is plenty. See the room notes.
Turner 95D as a mono crush mic (optional)	Low, in front of the kick, or out in the room at chest height	3-6 ft	At the kit	Into the 76-KT, all buttons in. Blend under the close mics. Its narrow range is the point.

Amps

Mic / channel	Where	Distance	Aim	Notes
Magnatone, SM57 (ch 8)	On the speaker cone, halfway between the dust cap (center) and the edge	1-2 in. off the grille	Straight at the cone	Toward the center = brighter and more bite; toward the edge = warmer. Off-axis (angled) = smoother top. The Magnatone's vibrato is in the amp, so the mic hears it; no need to add it later.
Marshall MG250DFX or Orange, SM58-style (ch 9)	Same as above	1-2 in.	At the cone, slightly off-axis	SM58-style mics are darker than a 57; start nearer the dust cap. For the 2x12 Marshall, pick the speaker that sounds best and mic only that one.
Bass amp, if mic'd instead of DI (ch 7)	On the cone, toward the edge	2-4 in.	At the cone	Combine with the DI for the best of both: mic for grit, DI for low end. Check mono for phase between them.
Blending two amps on group 6	Both mics as above			Both amps share group 6, so set each mic first, then balance on the console faders. Keep the two amps a few feet apart so each mic hears mostly its own.

Vocals, horn and acoustic

Mic / channel	Where	Distance	Aim	Notes
Lead vocal, Worker Bee (ch 10)	On a boom in front of the singer, capsule at mouth height or slightly above, with a pop filter	6-10 in. from the mouth, pop filter at 3-4 in.	At the mouth, singer slightly off to one side of the capsule's center	Closer for intimacy and more low end; back off for a shouted take. Hang a blanket or put the singer's back to the room's soft side: a condenser hears the whole room. LO CUT on at the console.
Vocal 2, Peavey (ch 15)	Handheld or on a stand	1-3 in. (dynamic mics like it close)	At the mouth, slightly off-axis to tame plosives	Dynamic mics reject the room, so they can sit near the drum kit when the band tracks together. Point its back at the kit.
Vocal 3, Turner 95D (ch 16)	On a stand	2-6 in.	At the mouth	Lo-fi by nature. Closer makes it fuller; it has no low end to over-emphasize. Sort its impedance first (Setup Map).
Trumpet, SM58 (ch 14)	In front of the bell, off-center	12-24 in. from the bell	Slightly off the bell's axis, not straight down it	Loudest source in the room. Straight into the bell is harsh and clips; off-axis and back a foot is round. The console trim will be low.
Acoustic guitar, Worker Bee (ch 10)	Where the neck meets the body	8-12 in.	At the 12th fret, angled slightly toward the sound hole	Never straight at the sound hole: that's all boom. For strummed parts pull back to 14-18 in.

The band tracking together: bleed control

- Order of volume in the room: drums, then amps, then trumpet, then voices. Put the loudest things farthest from the vocal mics, and point every cardioid mic's back at the kit.
- Guitar amps: face them away from the drums and away from each other, or put a blanket or a sofa cushion over the back and sides of an open-back combo. Even a chair with a coat on it between amp and kit helps.
- Vocals during a band take go on the dynamics (Peavey, SM58), not the Worker Bee: a condenser will record the drums as much as the voice. Overdub the lead vocal with the Worker Bee afterward.
- Bass through the DI needs no isolation at all; that's the main reason to use it for live takes.
- The room mics will hear everything. That's fine: in a small room they become "band mics" and glue the take. Check them in mono before committing.

Room-dependent decisions (what your measurements would change)

- **Where the kit sits.** Not in the exact center of the room and not against a wall; about a third of the way along the longer dimension, facing the long way, is the usual best spot. In a very small room, a corner-ish placement with the kick firing diagonally gives the room mics the most distance.
- **Room mic distance and height.** The defaults above assume a room of roughly 10-14 ft on its longer side. Smaller: bring the pair to 4-6 ft and lower, or use one mono room mic. Bigger: push back to 10-12 ft and raise to 6-7 ft. The ceiling height sets the maximum useful mic height: keep the capsules at least 18 in. below the ceiling.
- **Whether the room is live or dead.** Carpet and a low ceiling (like the control room in your photos) is dead: room mics will sound small, and the Turner crush mic becomes the better room sound. Hard floors and bare walls are live: room mics shine, but the vocal needs a blanket behind the singer.
- **Control room.** For mixing on the DIY monitors, the speakers should sit on the long wall's centerline with your head and the two speakers forming an equal triangle, and the first reflection points on the side walls (where a mirror held flat against the wall shows a speaker from your seat) get an absorber. The room's size sets the speaker spacing; with the console desk at about 4 ft wide, the clamp-mounted speakers land roughly 3.5-4 ft apart, so sit about 3.5-4 ft back.
- **What to send me:** length, width and ceiling height of the live room and the control room, and which walls are hard or soft. With those, the room-mic positions, the kit spot and the monitor triangle above become exact numbers instead of ranges.

Tascam 246: Setup and Recording

How the 246 Portastudio connects to the 24•8, how its own mixer is set, and the settings for recording to tape, bouncing, and getting four tracks back out. Everything here is for after the repair; it doesn't play tape yet. Control names follow the 246's panel.

What's on the 246 (so the rest makes sense)

- **Six input channels.** Channels 1-4 have a selector for MIC/LINE, OFF or REMIX (tape playback of that track). Channels 5-6 select MIC/LINE, OFF or LINE B (a pair of phono inputs on the back). Each channel has a trim with an overload LED, a two-band sweepable EQ (HI 1-8 kHz, LO 62 Hz-1.5 kHz, inner knob = cut/boost, outer = frequency), two post-fader effect sends, pan, and a fader. All inputs are unbalanced 1/4".
- **Assign buttons per channel: L/1, R/2, 3, 4.** A channel goes to tape track 1 or 2 through L/1 and R/2 with the pan (left = 1, right = 2), and to tracks 3 and 4 directly. At mixdown, L/1 and R/2 are the stereo left and right program busses.
- **Four RECORD FUNCTION buttons** arm the tracks; the LED flashes when armed and stays lit while recording. **Four big VU meters** show the tape tracks; two smaller ones show the stereo mix, the monitor mix or the effect sends depending on the meter select button.
- **Monitor section:** a level and pan for each of the four tape tracks (this is its own little cue mixer), a MONO button, PHONES level, and a CUE / PGM button. In PGM you hear the mix going to tape; in CUE you hear the tape tracks through the monitor section. The INSERT button is for punch-ins.
- **Back panel:** four TAPE OUT phonos (one per track, straight off the tape), stereo program outs, monitor outs, two effect send outs, LINE B inputs, six channel insert jacks, and two dbx noise-reduction slide switches (all tracks, or track 4 alone).
- **Under the cassette door:** the tape speed switch, 1-7/8 or 3-3/4 ips. Use 3-3/4 ips for everything you record here; it's cleaner and brighter. 1-7/8 exists only to play tapes made on slower machines. The pitch control (±12%) has its own on/off switch; leave it off unless you're doing it on purpose.

Connections

Direction	From	To	Cable	Notes
Console to tape	24•8 MAIN MIX OUT L and R	246 channels 5 and 6, selector on MIC/LINE (or LINE B phonos on the back)	1/4" TRS to 1/4" TS, 2	The 24•8's main outs are +4 dBu balanced; the 246 inputs are -10 dBV unbalanced. Expect to keep the console's main fader lower than usual and the 246 trims low.
Tape back to console	246 TAPE OUT 1-4 (phono), or the stereo program outs for a mixed pair	24•8 ch 23-24 line inputs	RCA to 1/4" TS, 2 (or 4 if you want all four tracks on their own console channels)	Ch 23-24 are the book's plan. For four separate tape tracks into the console at once, borrow ch 21-22 as well.
Into Reaper	24•8 ch 23-24 assigned to groups 7-8	UMC1820 inputs 7-8 (Reaper 7-8)	Already wired	Temporarily in place of the Reface CS. Record 23-24 in Reaper, then put the Reface back on 7-8.
Power	246 integral mains lead	Furman rear outlet 7 (the spare)		Low draw. It can share the Furman switch with the rack.

Levels: the one thing that matters most

- Set the 24•8 side first, as in the Gain Staging Guide: whatever you're sending to tape should be peaking near 0 on the console's L/R meters with the main fader at U.
- Then turn the 246's channel 5/6 trims up from zero until the big VU meters for the armed tracks peak around 0 VU on the loudest parts. The overload LED next to the trim must not flash. Cassette has very little headroom above 0 VU; +3 is the ceiling, and with dbx engaged even that starts to crunch.
- For the saturation sound you're after, run the meters into the red on drums and bass by a VU or two, then listen back. The crunch you hear is the tape, and it's permanent. Back off if cymbals turn to mush.
- Leave the 246's channel faders at the 0 mark for the tape-in path; use the trims for level, the same rule as on the console.
- If the tape comes back too quiet into the 24•8, raise the trims on ch 23-24, not the 246's program faders.

246 channel settings

246 channel	Used for	Trim	EQ HI (1-8 kHz)	EQ LO (62-1.5k)	Effect 1 / 2	Pan	Assign
5	24•8 main mix L	To 0 VU	Flat	Flat	Off	Hard L	L/1 (and 3 for a drum pass)
6	24•8 main mix R	Match ch 5	Flat	Flat	Off	Hard R	R/2 (and 4 for a drum pass)
1-4 on REMIX	Playback of tracks 1-4 at mixdown or bounce	To taste	Up to +3 dB at 5-6 kHz to put back the top tape loses	Cut 2-3 dB around 250 Hz if muddy	Off (the TSR-12 is on the console)	Spread L/R to taste	L/1 and R/2 = the stereo outs
1-4 on MIC/LINE	Recording straight into the 246 (4-track ideas away from the console)	To 0 VU	Flat to start	Flat	Off	Per assign rule	1 track each

The rule for EQ on the 246: none on the way in. The console's EQ already shaped the sound, and anything you add here is baked into the tape. Use its EQ only on the way back out, and lightly. The console's ch 23-24 EQ (HI shelf +1 dB) is the planned place for that.

Recording a pass to tape, step by step

Step	Do this	Watch for
1	Tape speed to 3-3/4 ips. dbx switch to the position for all four tracks (on). Pitch off. Clean tape, wound a few seconds past the leader.	The dbx LED. Decide dbx on or off before recording and never change it between recording and playback: a tape recorded with dbx on and played with it off sounds pumped and dull, and the reverse is hissy.
2	Press RECORD FUNCTION on the track(s) you're recording. Set the 246 to PGM so you hear what's going to tape.	The armed LEDs flash.
3	On the 24•8: keep ch 23-24 OUT of the L/R mix (their L/R buttons up). Otherwise the tape output feeds back into its own input.	If you hear a building howl or echo, this is why.
4	Set levels as above: console to 0, then 246 trims to 0 VU on the armed tracks.	Overload LEDs stay off.
5	Press RECORD and PLAY together. Record the pass. Stop. Zero return.	The counter: note where the take starts.
6	Playback check: 246 to CUE, raise that track's level in the monitor section, and listen on the 246's own headphone jack, or set ch 23-24 into L/R on the console and listen there.	Compare to the dry Reaper track. The difference is what tape did.
7	Capture into Reaper: assign ch 23-24 to groups 7-8, arm Reaper inputs 7-8, press play on the 246. Line the take up by ear or with the click at the start.	There is no sync between the 246 and Reaper. Record a short click or count-in at the head of every tape pass so lining up is easy.

Using all four tracks

- **Two stereo passes.** Pass one: console mix to tracks 1-2 (ch 5 to L/1, ch 6 to R/2). Pass two: a different mix or the same mix with heavier saturation to tracks 3-4 (ch 5 and 6 assigned to 3 and 4). Play all four back and blend them in the monitor section or on the console.
- **Drums-only crush.** Send only the drum groups to the console's L/R (mute everything else, or use the group masters), tape them to 3-4 with the meters hot, then capture that pair into Reaper as a parallel "tape drums" track to blend under the clean ones. This is the closest thing to the Tame Impala drum sound you can do without a reel-to-reel.
- **Bouncing inside the 246.** Put channels 1-3 on REMIX, assign them all to track 4, arm 4, record. Tracks 1-3 are now mixed onto 4 and can be re-used. Each bounce adds hiss and loses some top, so bounce once, not twice, and record the loudest sources last.
- **Four-track songwriting away from the console.** Channels 1-4 on MIC/LINE, one source per track, a mic straight into a 1/4" input. The 246 is a complete little studio on its own; dump the four TAPE OUTs into the console later.
- **Punch-ins.** Set the 246 to CUE with INSERT on, play up to the spot, press RECORD at the punch point; the monitor switches from tape to input at that moment. The optional footswitch jack does the same hands-free.

Things that go wrong

- Playback sounds dull and pumpy, or bright and hissy: dbx mismatch between record and play. Match the switch and try again.
- Howling or a slap echo when ch 23-24 are up: feedback loop. Take 23-24 out of L/R while recording to tape.
- Wow and flutter (pitch wobble): a worn belt or pinch roller, which is probably part of the repair. Also check that the pitch control is switched off.
- One channel of the pair is quiet: a trim mismatch between 246 ch 5 and 6, or the console's main L/R fader isn't the stereo pair you think (it is on the 8•Bus; check the sub-master assignment).
- The tape dropouts on track 4: the dbx switch for track 4 is separate for a reason. Try it off for that track if you're recording a sync tone or a click there.
- Head cleaning: cotton swab and isopropyl on the heads, capstan and pinch roller every few hours of use. Dirty heads lose high end before anything else.

Reaper Setup

Getting Reaper on the PC to see all 16 inputs and 16 outputs through the UMC1820, with a track template that matches the console channel map, playback going back to the tape returns, MIDI for the pads, and recording settings that leave headroom.

1. Driver and audio device (once)

- Install Behringer's UMC driver for Windows from Behringer's site before plugging the UMC1820 in. Use the ASIO driver, not Windows' built-in WASAPI or DirectSound: ASIO is the only mode that exposes all 18 inputs and 20 outputs and gives low latency.
- In Reaper: Options > Preferences > Audio > Device. Audio system: **ASIO**. ASIO driver: **UMC ASIO Driver** (the name on your system may say BEHRINGER UMC). Enable inputs: first **1**, last **16**. Enable outputs: first **1**, last **18**.
- Sample rate: **48000** (or 44100). Never 88.2 or 96 kHz: ADAT drops to 4 channels and inputs 13-16 vanish. Set the same rate on the ADA8200's switch. Block size: start at **256**; go to 128 if headphone latency bothers performers, 512 if you hear clicks.
- Click ASIO Configuration to open the UMC control panel and confirm the sample rate and clock there too. The UMC1820's inputs 9-16 are the ADAT input (the ADA8200); 17-18 are S/PDIF, which you aren't using.
- Preferences > Project > Media: recording format **WAV, 24-bit**. 24-bit is what gives you room to record at -12 dBFS without losing anything.

2. Name the inputs and outputs (once)

- Preferences > Audio > Device > Input channel name aliasing: name them so the track-input menu reads like the console map. Inputs 1-8 are the groups; 9-16 are the ADA8200.
- Suggested names: 1 Kick (grp1), 2 Snare (grp2), 3 Drums L (grp3), 4 Drums R (grp4), 5 Bass (grp5), 6 Guitars (grp6), 7 Reface L / Tape L (grp7), 8 Reface R / Tape R (grp8), 9 Lead vocal (ADA1), 10 Wireless (ADA2), 11 Trumpet (ADA3), 12 Vocal 2 (ADA4), 13 Vocal 3 (ADA5), 14 SH-201 L (ADA6), 15 SH-201 R (ADA7), 16 Drum machine (ADA8).
- Output aliasing: 1-2 Tape ret 1-2 (UMC main), 3-8 Tape ret 3-8, 9-10 unused (UMC 9-10), 11-18 Tape ret 9-16 (ADAT to the ADA8200). Reaper numbers the UMC1820's outputs 1-10 as the analog jacks and 11-18 as ADAT; the console's tape returns 9-16 are therefore Reaper outputs 11-18.

3. The track template (matches the console map)

Reaper track	Input	Record arm default	Output (playback)	Console channel it plays back on
1 Kick	Mono 1	Armed in the drum template	Out 1	Tape return 1
2 Snare	Mono 2	Armed	Out 2	Tape return 2
3-4 Drums (toms + rooms)	Stereo 3/4	Armed	Outs 3/4	Tape returns 3-4
5 Bass	Mono 5	Armed	Out 5	Tape return 5
6 Guitars	Mono 6	Armed	Out 6	Tape return 6
7-8 Reface CS (or Tascam capture)	Stereo 7/8	Off until used	Outs 7/8	Tape returns 7-8
9 Lead vocal	Mono 9	Armed	Out 11	Tape return 9
10 Wireless	Mono 10	Off until used	Out 12	Tape return 10
11 Trumpet	Mono 11	Off until used	Out 13	Tape return 11
12 Vocal 2	Mono 12	Off until used	Out 14	Tape return 12
13 Vocal 3	Mono 13	Off until used	Out 15	Tape return 13
14-15 SH-201	Stereo 14/15	Off until used	Outs 16/17	Tape returns 14-15
16 Drum machine	Mono 16	Off until used	Out 18	Tape return 16
17 E-drums (MIDI)	MIDI: UMC1820 MIDI in, all channels	Armed when the pad is in use	Out 18 (shares the drum machine's return) or the main 1/2	Tape return 16, or 1-2

Reaper track	Input	Record arm default	Output (playback)	Console channel it plays back on
18 SamplePad (MIDI)	MIDI: SamplePad USB, all channels	Off until used	Main 1/2	Tape returns 1-2
MASTER			Outs 1/2	Tape returns 1-2 (MIX-B)

Build this once, then File > Project templates > Save project as template, named "Toad Studio tracking". Every new session starts from it. Set each track's input with the small IN button on the track panel, and its output in the track's Routing window: untick "Master send" and add a hardware output to the out listed above. The MASTER goes to outs 1/2 so a quick mix comes back on tape returns 1-2 when you press MIX-B.

4. Playback back to the console

- Two ways to hear Reaper, pick per session. **Per-track playback:** every track to its own hardware out as in the table, so each Reaper track comes back on its own console tape return and you mix on the console's MIX-B section (or flip the channel to TAPE). That's the 16-track overdub setup.
- **Mix playback:** tracks send to the master only, master to outs 1/2, and the whole mix comes back on tape returns 1-2. Simpler for writing and for mixing in Reaper; press MIX-B on the console and you hear it.
- The UMC1820's front MAIN OUT knob sets outs 1-2 only. Keep it all the way up; the console's MIX-B level is your volume.
- Don't route a track to the same hardware out as the input it's recording from while the console channel is in the L/R mix: that's a feedback loop through the console. The per-track plan avoids it because recording comes from group/direct outs and playback goes to tape returns, which are separate paths.

5. Recording settings

- Preferences > Audio > Recording: **input monitoring off** on all audio tracks (the track panel's monitor button). You're hearing yourself through the console, with zero latency; Reaper's monitoring would add a delayed copy. Turn Reaper monitoring on only for the MIDI tracks, which need it to hear the virtual drums.
- Record mode (Options > Record mode): **Normal** for tracking. Use **Time selection auto punch** for punch-ins; it starts and stops recording at the selection edges.
- Levels: the whole book aims for peaks around -12 to -6 dBFS on the Reaper meters. At 24-bit that's plenty. If a track's meter is touching the top, fix it at the console trim, not with Reaper's input gain (there isn't one; the fader is playback only).
- Pre-roll and count-in: Preferences > Audio > Recording > "Count-in before recording": 2 bars, and the metronome on during count-in only. The count-in through the headphone mix is also the click you'll use to line up Tascam passes.
- Project settings (Alt+Enter): BPM and time signature per song; **Timebase for items: Time**, so a later tempo change doesn't stretch recorded audio.
- Autosave: Preferences > Project > "Save project every 5 minutes" and "when recording stops". Record files into a dated project folder on the PC, not a USB drive.

6. MIDI for the e-drums and SamplePad

- Preferences > Audio > MIDI Devices: enable **UMC1820 MIDI In** (the e-drum module) and the **SamplePad 4** USB MIDI port as inputs. Enable UMC1820 MIDI Out only if you want Reaper to drive the module's sounds.
- On the MIDI tracks, insert **ReaSamplOmatic5000** or a drum instrument plugin (the free MT Power Drum Kit works well for the one-pad-at-a-time method) and arm with monitoring on. Record the pad hits as MIDI notes, then move or re-pitch notes to any drum sound afterwards.
- To record the module's own sounds as audio instead, that's the PV6 path on console channels 19-20, which is monitor-only by design. Use the MIDI path for anything you want to keep.

7. Mixing in Reaper: the plugins that replace outboard

- **ReaComp** on the drum bus and master does the clean bus compression the dbx units would. **ReaEQ** for anything beyond the console's broad strokes. **ReaDelay** and **ReaVerb** stand in for the aux-2 delay until you buy one.
- For the Tame Impala crush in the box: a free 1176-style plugin on the room-mic track, or ReaComp with ratio 20:1, fast attack, fast release, pushed 10-15 dB, blended under the clean drums. Not as good as the real 76-KT, but close for demos.
- Render (File > Render): WAV 24-bit for masters, MP3 320 for sharing. Render to a "Mixes" folder inside the project folder with the date in the filename.

Troubleshooting

- Reaper shows only 8 inputs: audio system isn't ASIO, or last input is set to 8. Both are in Preferences > Audio > Device.
- Inputs 9-16 are silent or clicky: the ADA8200 and UMC1820 aren't on the same clock, or the sample rates differ. See the Setup Map's clock notes; the ADA8200's SYNC light must read LOCKED.
- Doubled, slightly delayed sound in headphones while recording: Reaper's input monitoring is on. Turn it off on the audio tracks.
- Playback is heard only on tape returns 1-2: the tracks are sending to the master instead of their own hardware outs. Check each track's Routing window.
- Crackles when the screen is busy: raise the block size one step, and set Windows' power plan to High performance. USB hubs cause this too; the UMC1820 should be on its own port on the PC.
- Reaper can't find the files after a session: the project was saved to a different folder than the media. Use File > Save project as with "Copy all media into project directory" ticked once, and the template's folder setting after that.

FL Studio Setup

The same studio in FL Studio instead of Reaper. The driver, sample rate, clock and all the wiring are identical to the Reaper page; what's different is how FL takes in audio, which runs through the Mixer rather than tracks. Recording external audio needs FL Studio Producer Edition or higher. Control names follow the FL Studio manual.

What's the same as Reaper (don't redo it)

- Behringer's UMC ASIO driver, 48 kHz, 24-bit, the same block size, the ADA8200 clocked to the UMC1820: all exactly as in the Reaper page, sections 1 and the clock notes in the Setup Map.
- Inputs 1-8 are the console groups, 9-16 are the ADA8200 (direct outs). Outputs 1-10 are the UMC1820's analog jacks (tape returns 1-8 on outputs 1-8), outputs 11-18 are the ADAT outs (tape returns 9-16).
- You still monitor through the console, not the computer. Latency and feedback rules are the same.

1. Audio settings (once)

- Options > Audio Settings (F10). Device: the Behringer UMC ASIO driver. The manual's own advice: use the manufacturer's ASIO driver; "FL Studio ASIO" is a compatibility fallback that adds latency and jitter, and ASIO4ALL is only for combining several devices. One interface means the Behringer driver.
- Sample rate 48000. Buffer length: start around 256 samples (about 5 ms); larger if you hear underruns, which FL shows as a counter in the Audio Settings window.
- Mixer menu > Disk recording: tick **Auto-unarm** so armed inserts drop out after a take, and **Latency compensation** so recorded clips line up with the Playlist.
- Options > Project general settings: default recording folder inside a dated project folder on the PC, same habit as Reaper.

2. Map the 16 inputs to Mixer inserts (once, then save as template)

- Open the Mixer (F9). On Insert 1, right-click the **Input** menu (top right of the insert, with the Recording Pickup selector in front of it) and choose **Auto-map**. FL assigns each device input to a successive insert: input 1 to Insert 1, through input 16 to Insert 16. It also disables those inserts' sends to the Master so the live input can't feed back; you see meters but hear nothing, which is correct because you're listening through the console.
- Mono vs stereo: the Input menu has a stereo list on top and a mono list below. For every mono source (kick, snare, bass, vocals, trumpet, drum machine) pick the mono entry so it feeds both sides. Use the stereo pairs only for inputs 3/4 (drums), 7/8 (Reface) and 14/15 (SH-201). Auto-map picks stereo pairs, so fix the mono ones by hand afterwards.
- Recording Pickup selector on each insert: **EXT** (external), the manual's recommendation, so the recording is the raw input before any insert effects.
- Name and color the inserts to match the console map (right-click the insert name, F2): 1 Kick, 2 Snare, 3-4 Drums, 5 Bass, 6 Guitars, 7-8 Reface / Tape, 9 Lead vocal, 10 Wireless, 11 Trumpet, 12 Vocal 2, 13 Vocal 3, 14-15 SH-201, 16 Drum machine.
- Link each insert to a Playlist track so takes land in a predictable place: right-click a Playlist track header > Track mode > Audio track > pick the insert. Do this for all 16, in order.
- File > Save as template (or save to Templates folder) as "Toad Studio tracking".

3. Playback back to the console

Mixer insert	Output (insert's OUT menu)	Console tape return	Note
1-8 (groups)	UMC outputs 1-8, one per insert	Tape returns 1-8	Pair inserts 3/4 and 7/8 as stereo outs 3-4 and 7-8.
9-16 (ADA8200)	UMC outputs 11-18, one per insert	Tape returns 9-16	FL numbers the ADAT outs 11-18 exactly as Reaper does.
Master	UMC outputs 1-2	Tape returns 1-2	The whole mix comes back on MIX-B when you only want a stereo return.

Each insert's output is chosen in its OUT menu, next to the Input menu. By default everything goes to the Master; for 16-track overdubbing give each insert its own hardware output as above and leave the Master on 1-2. Keep the UMC1820's MAIN OUT knob all the way up; MIX-B level on the console is the volume.

4. Recording a take

- Arm the inserts you're recording: click the **disk icon** at the bottom of each insert (selecting an input arms it automatically; check it's lit).
- Right-click the transport **Record** button and make sure **Audio** is ticked in the recording filter. This is the step everyone misses: with Audio unticked, FL records knob moves and notes but no sound.
- Left-click Record so it lights orange, then press Play. Recording runs until Stop. In Song mode with Auto-create audio clip on, the takes appear as Audio Clips on the linked Playlist tracks.
- Count-in: the one-click recording mic icon in the toolbar gives a 4-beat count-in and records straight into the Playlist; use it for quick takes. For the count-in click through the headphones, turn the metronome on and set it to count-in only.
- Levels: the same -12 to -6 dBFS target on the insert meters. Set it at the console trim; the insert fader is playback only.
- **Monitor external input: Off** on every audio insert (right-click the insert's input area). You hear yourself through the console; FL's monitoring would add a delayed copy. If a take comes back slightly early or late against the click, use the Audio input delay compensation control in the Mixer menu.

5. MIDI for the e-drums and SamplePad

- Options > MIDI Settings: enable the UMC1820 MIDI In (e-drum module) and the SamplePad 4 USB port as inputs. Set each to a different controller type only if FL asks; generic is fine.
- Add FPC (FL's drum sampler) or a drum VST on a Channel Rack channel, route it to a free Mixer insert (17 or above), and record the pad hits as a Pattern. One pad can be any drum; move notes in the Piano roll to re-assign them.
- FL records MIDI into Patterns by default, so for the pads use Pattern mode or make sure the Notes box is ticked in the recording filter.

6. Mixing in FL: the plugins that replace outboard

- Fruity Limiter in compressor mode, or Maximus, for bus compression; Fruity Parametric EQ 2 for surgery; Fruity Delay 3 for the aux-2 delay stand-in; Fruity Reeverb 2 for reverb.
- The room-mic crush: Fruity Limiter with ratio high, fast attack and release, 10-15 dB of gain reduction, on the room-mic insert, blended under the clean drums.
- Export: File > Export > WAV, 24-bit, for masters; MP3 320 for sharing, to a "Mixes" folder inside the project folder.

Reaper or FL Studio for this studio?

- Either works with everything in this book. The console does the tracking work in both cases; the computer is a 16-track tape machine.
- Reaper's track-per-input model maps onto the console more directly and its per-track hardware outputs are quicker to set up. FL Studio's strengths are the Channel Rack and Piano roll for the e-drums, SamplePad and drum machine programming.
- A reasonable split: track the band in Reaper, program beats and synth parts in FL, and move audio between them as WAV files. Both read the same 24-bit 48 kHz files, so nothing is lost.

Troubleshooting

- Nothing records: the recording filter's Audio box is unticked, or no insert has its disk icon lit, or the input is set on the insert but it's a stereo entry for a mono mic (you'll get one side only).
- FL sees only 2 inputs or none: Audio Settings isn't on the Behringer ASIO driver.
- A doubled, delayed sound while playing: Monitor external input is on. Turn it off; you're monitoring through the console.
- Takes land on the wrong Playlist track: the insert isn't linked to a Playlist track; set Track mode > Audio track on the track you want.
- Clicks on inputs 9-16 only: clock. The ADA8200's SYNC light must read LOCKED (Setup Map).
- Everything in the Playlist is a quarter-beat late: Disk recording > Latency compensation is off.

Pro Tools Setup

The same studio in Pro Tools. Driver, sample rate, clock and wiring are identical to the Reaper page. Pro Tools adds one step the others don't have, the I/O Setup window, where every input and output has to exist as a named path before a track can use it. Control names follow Avid's current Pro Tools.

What's the same as Reaper (don't redo it)

- Behringer's UMC ASIO driver on Windows, 48 kHz, 24-bit, the ADA8200 clocked to the UMC1820: all as in the Reaper page and the Setup Map's clock notes.
- Inputs 1-8 are the console groups, 9-16 are the ADA8200. Outputs 1-10 are the UMC1820's analog jacks (tape returns 1-8 on 1-8), outputs 11-18 are the ADAT outs (tape returns 9-16).
- You monitor through the console, not the computer.
- Tier check before you start: Pro Tools Intro is limited to 8 audio tracks total, so it can't run this 16-input studio. Pro Tools Artist (16 simultaneous record inputs) is the minimum; Studio removes the limits. Check the current Avid tier table, since these numbers change.

1. Session and Playback Engine (once)

- Dashboard > Create: name it, **WAV, 24-bit, 48 kHz**, I/O Settings: Last Used. Pro Tools fixes the sample rate per session, so pick 48 kHz here and never change the UMC1820 or ADA8200 afterwards.
- Setup > Playback Engine. Playback Engine: the **Behringer UMC ASIO** driver (Windows lists ASIO drivers here). H/W Buffer Size: **256** for tracking; raise to 1024 when mixing with lots of plugins. Voices: leave at the default. Holding N while Pro Tools starts opens this window if the wrong device is stuck.
- Setup > Session (Ctrl+2): Clock Source should be **Internal** on the UMC1820 side. The ADA8200 follows it over ADAT (SYNC switch to ADAT IN, LOCKED light on).

2. I/O Setup (once, then export the settings)

- Setup > I/O (Ctrl+Shift+I). This is the step that trips people up: Pro Tools only offers a track the paths defined here.
- **Input tab:** click **Default** to let Pro Tools create a path per physical input, then rename them to the console map. Make inputs 3-4, 7-8 and 14-15 stereo paths and leave the rest mono. Click a name to rename: Kick, Snare, Drums (3-4), Bass, Guitars, Reface/Tape (7-8), Lead vocal (ADA1), Wireless (ADA2), Trumpet (ADA3), Vocal 2 (ADA4), Vocal 3 (ADA5), SH-201 (ADA6-7), Drum machine (ADA8).
- **Output tab:** Default, then rename. Outputs 1-2 are the UMC1820's main jacks (tape returns 1-2); make them a stereo path called "Main / Tape 1-2". Outputs 3-8 mono paths "Tape 3" through "Tape 8" (or stereo pairs for 3-4 and 7-8). Outputs 11-18 are the ADAT outs: name them "Tape 9" through "Tape 16". Outputs 9-10 are the UMC1820's unused analog pair; leave them.
- **Bus tab:** leave the defaults. Set the **Monitor Path** (bottom of the Output tab) to Main / Tape 1-2.
- Click **Export Settings** and save as "Toad Studio.pio" so a new session can Import Settings instead of redoing this.

3. The session template

Track	Input path	Output path	Console tape return
Kick (mono)	Kick (in 1)	Tape 1	1
Snare (mono)	Snare (in 2)	Tape 2	2
Drums (stereo)	Drums (in 3-4)	Tape 3-4	3-4
Bass (mono)	Bass (in 5)	Tape 5	5
Guitars (mono)	Guitars (in 6)	Tape 6	6
Reface / Tape capture (stereo)	Reface/Tape (in 7-8)	Tape 7-8	7-8
Lead vocal (mono)	Lead vocal (in 9)	Tape 9 (out 11)	9
Wireless (mono)	Wireless (in 10)	Tape 10 (out 12)	10
Trumpet (mono)	Trumpet (in 11)	Tape 11 (out 13)	11
Vocal 2 (mono)	Vocal 2 (in 12)	Tape 12 (out 14)	12
Vocal 3 (mono)	Vocal 3 (in 13)	Tape 13 (out 15)	13
SH-201 (stereo)	SH-201 (in 14-15)	Tape 14-15 (outs 16-17)	14-15

Track	Input path	Output path	Console tape return
Drum machine (mono)	Drum machine (in 16)	Tape 16 (out 18)	16
E-drums (Instrument track)	MIDI: UMC1820 MIDI In, all	Main / Tape 1-2	1-2
SamplePad (Instrument track)	MIDI: SamplePad USB, all	Main / Tape 1-2	1-2
Master Fader		Main / Tape 1-2	1-2 (MIX-B)

Track > New (Ctrl+Shift+N): make the mono and stereo audio tracks in one go, timebase Samples. Set each track's Input and Output selectors in the Mix window's I/O section (the two buttons above the fader). Then File > Save as Template, named "Toad Studio tracking", so Dashboard offers it under Templates. With each track on its own output, every Pro Tools track comes back on its own console tape return; switch all outputs to Main / Tape 1-2 when you only want a stereo return on MIX-B.

4. Recording

- Record-enable tracks with the **R** button on each track (Alt-click arms every track). It blinks red when armed and stays solid once the transport is record-armed.
- Monitoring: you listen through the console, so Pro Tools shouldn't also feed you the live input. Leave **Options > Auto Input Monitoring** on (its default) and **Input Only Monitoring** off, and keep the console's tape-return channels for the tracks you're recording out of the L/R mix during the take. Bring them back in for playback. If you ever do monitor through Pro Tools, **Options > Low Latency Monitoring** bypasses plugins on armed tracks to keep the delay small.
- Transport: record button, then Play (or Ctrl+Space). For punch-ins, Options > **QuickPunch**, which lets you punch in and out on the fly with Ctrl+Space while rolling.
- Count-off: Setup > Click/Countoff Options: 2 bars, countoff only during record. Add a click track with Track > Create Click Track; its output can go to Main / Tape 1-2 so it reaches the headphones through MIX-B, and is also the click you'll use to line up Tascam passes.
- Levels: the same -12 to -6 dBFS target on the track meters. Pro Tools track meters are post-fader by default; Setup > Preferences > Metering lets you switch them to pre-fader, which is more useful while tracking.
- Setup > Preferences > Operation: Auto Backup on, every 5 minutes, keep 10. Sessions and audio into a dated session folder on the PC.

5. MIDI for the e-drums and SamplePad

- Setup > MIDI > Input Devices: tick the UMC1820 MIDI In and the SamplePad 4. Create two Instrument tracks, set each one's MIDI input to its device (all channels), and insert a drum instrument (Pro Tools ships with Boom and Xpand!2; a free third-party kit works too). Record-enable and record the pad hits as MIDI; re-assign notes to any drum afterwards in the MIDI Editor.

6. Mixing: the plugins that replace outboard

- Dyn3 Compressor/Limiter on the drum bus and master; EQ3 7-Band for surgery; Mod Delay III for the aux-2 delay stand-in; D-Verb for reverb. All ship with Pro Tools.
- The room-mic crush: BF-76 (Avid's included 1176 model) on the room-mic track, all four ratio buttons in, input pushed hard, blended under the clean drums. This is the closest in-the-box match to the 76-KT you're buying.
- Bounce: File > Bounce Mix, WAV 24-bit for masters, MP3 for sharing, to a "Mixes" folder in the session folder.

Troubleshooting

- A track's input menu only shows a few inputs: I/O Setup hasn't been defaulted or the paths aren't active. Setup > I/O > Input tab > Default.
- No sound from inputs 9-16 or clicks on them: clock. Session Setup clock on Internal, ADA8200 SYNC on ADAT IN and LOCKED.
- Audio outputs aren't selectable above 2: Playback Engine is on the wrong device, or the Output tab needs Default.
- Doubled, delayed sound while recording: Pro Tools' monitoring is reaching the console through the tape returns while the live channel is also in the mix. Take the tape-return channels out of L/R while tracking.
- "Session sample rate doesn't match": the UMC1820 is at a different rate than the session. Match it in the UMC control panel and the ADA8200 switch, then reopen the session.
- Pro Tools won't open the session's audio: it was moved without the Audio Files folder. Always move the whole session folder.

GarageBand Setup

The same studio in GarageBand. Two things to know first: GarageBand runs on Mac and iPad only, so this page applies if a Mac joins the studio alongside the Windows PC; and GarageBand is built for small sessions, so it works for overdubs and demos but can't do the full 16-input band tracking the other pages describe. Control names follow Apple's GarageBand User Guide.

What's different about GarageBand

- **Simultaneous recording is limited.** Apple's guide says you can record to more than one audio track at a time but doesn't publish a cap; in practice GarageBand users report a ceiling of around 8 armed tracks. Treat it as an 8-input recorder: the console's groups 1-8 into the UMC1820, with the ADA8200's 8 extra inputs for overdubs one at a time. Logic Pro, GarageBand's big brother, removes the limit and opens GarageBand projects directly.
- **No ASIO.** On a Mac the UMC1820 and ADA8200 are class-compliant: no driver to install. The Mac's Core Audio sees the UMC1820 with all 18 inputs and 20 outputs.
- **Outputs are stereo only.** GarageBand sends everything to one stereo output pair. There is no per-track hardware output, so there is no 16-track playback to the tape returns. Playback comes back on tape returns 1-2 and you press MIX-B on the console. For per-track returns you need Logic, Reaper (which also runs on Mac), or Pro Tools.
- **Projects are fixed at 44.1 kHz** in GarageBand's settings (Preferences > Audio/MIDI > Sample rate offers 44.1 or 48 in recent versions; older versions are 44.1 only). Whichever you choose, set the UMC1820 and ADA8200 to match.

1. Preferences (once)

- GarageBand > Settings (Preferences) > Audio/MIDI. Output Device: **UMC1820**. Input Device: **UMC1820**. Sample rate 48 kHz if offered, otherwise 44.1 kHz, and set the ADA8200's switch and the UMC1820 to the same.
- In the same pane, set the buffer (I/O Buffer Size) to Medium or Small for tracking. GarageBand hides latency controls behind these presets.
- Audio MIDI Setup (in Applications > Utilities): confirm the UMC1820 shows 18 ins and 20 outs and the right sample rate. This app is where the Mac's clock and rate are set; GarageBand follows it.
- Settings > Advanced: tick **Show advanced tools**. Without it, Record Enable buttons and per-track input selection don't appear.

2. Make the tracks and set their inputs

- Track > New Track > Audio (mic icon), **Input: Input 1** for a mono source or **Input 1 + 2** for a stereo pair. Repeat per source; GarageBand asks for the input as you create each track, which is the quickest way to map the console channels.
- To change an input later: select the track, open the Smart Controls (B), and use the **Input** menu in the recording settings area, or double-click the track header for the track's inspector.
- Show the Record Enable buttons: Track > **Configure Track Header** (or Option-T), tick **Record Enable**. Apple's guide: the Record Enable button must be visible on each track you want to record to.
- Name and color the tracks to match the console map, and save the project as a template by keeping a blank copy called "Toad Studio tracking" (GarageBand has no template feature; File > Save As and duplicate it per session).

3. The track layout

GarageBand track	Input (UMC1820)	Console source	Armable at once?
Kick	Input 1	Group 1	Yes
Snare	Input 2	Group 2	Yes
Drums (stereo)	Input 3 + 4	Groups 3-4	Yes
Bass	Input 5	Group 5	Yes
Guitars	Input 6	Group 6	Yes
Reface / Tape (stereo)	Input 7 + 8	Groups 7-8	Yes (this is the 8-input live set)
Lead vocal	Input 9	ADA8200 1 (ch 10 direct out)	As an overdub, or swap into the live set
Wireless	Input 10	ADA8200 2	Overdub
Trumpet	Input 11	ADA8200 3	Overdub
Vocal 2	Input 12	ADA8200 4	Overdub

GarageBand track	Input (UMC1820)	Console source	Armable at once?
Vocal 3	Input 13	ADA8200 5	Overdub
SH-201 (stereo)	Input 14 + 15	ADA8200 6-7	Overdub
Drum machine	Input 16	ADA8200 8	Overdub
E-drums (Software Instrument)	MIDI: UMC1820 MIDI In	Pad	Yes
SamplePad (Software Instrument)	MIDI: SamplePad USB	Pad	Yes

For a live band take, arm the first eight (the group inputs) and record the drums, bass, guitars and keys together; the lead vocal goes down in the same pass only if you move it onto a group (assign ch 10 to group 7 or 8 instead of the Reface for that song). Everything on the ADA8200 inputs is then overdubbed one or two at a time, which is how most GarageBand records get made anyway.

4. Recording

• Arm the tracks (Record Enable buttons). Apple's guide notes that record-enabling a track also lets you monitor the input in GarageBand; turn each armed track's **monitoring** off (the small speaker icon next to Record Enable, or in Smart Controls) because you're hearing yourself through the console and would otherwise get a delayed double.
• Set the metronome and count-in (the metronome and count-in buttons in the control bar; count-in length in Settings > Metronome). The count-in is also your click for lining up Tascam passes.
• Press Record (R), play, press Stop. Multiple takes on one track go into a take folder; click the take number to choose.
• Levels: the same -12 to -6 dBFS target on the track meters, set at the console trim. GarageBand's track volume is playback only; its recording level control (in Smart Controls) stays at 0 because the UMC1820's gain is at minimum and the console is doing the work.
• Punch-ins: Record > Autopunch (or the Autopunch button in the control bar) and drag the red punch range in the ruler.

5. Playback back to the console

• GarageBand's stereo output goes to UMC1820 outputs 1-2, which are tape returns 1-2. Press MIX-B on the console and raise MIX-B LEVEL. That's it; there is no per-track return.
• For a performer to hear a backing track and themselves, the console does the blend: MIX-B (GarageBand) plus the live channel on the aux 3/4 performer mix or on MONITOR.
• The UMC1820's front MAIN OUT knob sets outputs 1-2. Keep it all the way up.

6. MIDI and the built-in instruments

• The UMC1820's MIDI in and the SamplePad's USB port appear automatically in GarageBand once the Mac sees them. Create a Software Instrument track, pick a Drum Kit, and play the pad; GarageBand records the MIDI. Drummer tracks and Apple's drum kits are a real strength here for the one-pad-at-a-time method.

7. Mixing: the plugins that replace outboard

• Compressor and Channel EQ on every track (GarageBand's are Logic's, trimmed down). Delay and Reverb on the two master effect sends stand in for the TSR-12 and the aux-2 delay.
• For the room-mic crush: Compressor set to the Vintage FET model, ratio high, fast attack and release, pushed hard, on the room-mic track under the clean drums.
• Share > Export Song to Disk: WAV 24-bit for masters, AAC or MP3 for sharing.

When to move up

• You'll feel GarageBand's limits the first time the whole band wants to record together with vocals, or the first time you want each track back on its own console channel. Logic Pro is the natural step: same files, same look, no track limits, per-track outputs, and it opens every GarageBand project you've made. Reaper also runs on the Mac and would let the PC and Mac share one workflow.

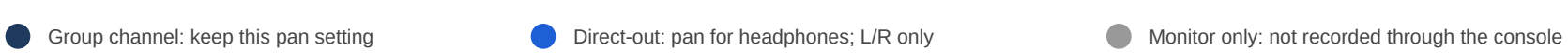
Troubleshooting

- Only inputs 1-2 appear: Input Device isn't the UMC1820, or Show advanced tools is off.
- Tracks can't all be armed: you've hit the simultaneous-record ceiling. Record in two passes, or move to Logic or Reaper.
- Doubled, delayed sound: track monitoring is on. Turn it off; the console is your monitor.
- Clicks on inputs 9-16: sample rate mismatch between the project, the UMC1820 and the ADA8200. Set all three the same in Audio MIDI Setup and on the ADA8200.
- The project plays back at the wrong pitch after changing the rate: GarageBand resampled. Keep one rate per project from the start.

Panning Guide, Part 1: Console Pan Knobs (Tracking)

Dot = 24•8 pan knob position. Group channels: pan picks the group, keep it fixed. Direct-out channels: pan only changes the headphones.

Ch	Source	L C R	Records to
1	Kick		**Group 1**
2	Snare		**Group 2**
3	Rack tom (SM58-style)		**Groups 3-4**
4	Floor tom (SM58-style)		**Groups 3-4**
5	Room L		**Groups 3-4**
6	Room R		**Groups 3-4**
7	Bass amp (DI out or mic)		**Group 5**
8	Magnatone (SM57)		**Group 6**
9	Marshall or Orange (SM58-style)		**Group 6**
10	Lead vocal (Worker Bee)		Direct out > ADA8200 in 1
11	ATW-R3210 wireless		Direct out > ADA8200 in 2
12	Reface CS L		**Group 7**
13	Reface CS R		**Group 8**
14	Trumpet (SM58)		Direct out > ADA8200 in 3
15	Vocal 2 (Peavey)		Direct out > ADA8200 in 4
16	Vocal 3 (Turner 95D)		Direct out > ADA8200 in 5
17	Roland SH-201 L		Direct out > ADA8200 in 6
18	Roland SH-201 R		Direct out > ADA8200 in 7
19	Peavey PV6 main out L		Monitor only
20	Peavey PV6 main out R		Monitor only
21	1970s Roland drum machine		Direct out > ADA8200 in 8
22	Open (drum machine R if stereo)		
23	Tascam 246 line out L		Monitor; groups 7-8 to capture
24	Tascam 246 line out R		Monitor; groups 7-8 to capture



Lopsided headphones? Press MONO or use the AUX SEND 3/4 performer mix. Neither changes the recording. Set final panning in Reaper.

Panning Guide, Part 2: Final Mix in Reaper

Suggested starting points for mixdown. Console pan doesn't carry over; set these on Reaper's tracks.

Reaper	Track	L C R	Why
1	Kick		Center: anchors the low end.
2	Snare		Center (or just off, to taste).
3-4	Toms + rooms (stereo)		Stereo track at full width. Rack tom ends up left of center, floor tom right, matching the console pans.
5	Bass		Center: low end stays mono.
6	Guitars (Magnatone + Marshall/Orange)		Slightly left; for wide guitars, double-track and pan the second take right.
7-8	Reface CS (stereo)		Stereo track at about 70% width so it doesn't crowd the guitars.
9	Lead vocal		Center, always.
10	Wireless		Center if it's a lead part; off-center if it's an instrument.
11	Trumpet		Right, opposite the guitar track.
12	Vocal 2		Harmony left.
13	Vocal 3		Harmony right, mirroring vocal 2.
14-15	Roland SH-201 (stereo)		Stereo pair: hard L/R for a wide pad, or narrow it if it fights the Reface.
16	1970s drum machine		Center, or slightly off if it's a percussion layer.

Mono track: pan position Stereo track: width

Rules of thumb: kick, snare, bass and lead vocal stay centered. Balance the sides: what goes left, something similar goes right.

Check the mix in mono (Reaper master mono button or the console's MONO): nothing important should disappear.

Headphones exaggerate width; once the monitors are built, re-check panning on them.

Alternate Setups

Other ways to wire the same studio. The Setup Map is the main plan and stays as it is; everything in this chapter is an option to try, with what changes, what it costs you, and how to switch back. New variations get added here as Alternate B, C and so on.

How to use this chapter

- Each alternate lists only what's **different** from the main plan. Anything not mentioned (rack, power, playback to the tape returns, clock, aux sends, monitoring outputs) stays exactly as the rest of the book describes.
- Every alternate here can be tried and undone in an afternoon by moving snake tails at the back of the console. Nothing needs buying.

Alternates can be combined: for example, Alternate A's recording with Alternate B's playback, which is what the connection diagram at the end of this chapter shows.

Alternate A: all direct outs

Recording skips the eight groups. Each of channels 1-16 sends its own direct out to its own Reaper input, so channel 1 = Reaper track 1 through channel 16 = Reaper track 16, and each track plays back on the same channel's tape return. This is a standard way to track on an in-line console like the 24•8: every source gets its own track, and the balance between them is decided in the mix instead of at tracking.

Alternate A channel map

The sources stay on the same channels as the main map; only what the snakes plug into changes. Pans now affect only what you hear, so they can be set naturally (kick and bass centered) instead of hard left/right to pick a group.

Ch	Source	Records to	Pan (headphones only)
1	Kick	Direct out → UMC1820 in 1 (Reaper 1)	Center
2	Snare	Direct out → UMC1820 in 2 (Reaper 2)	Center or slightly R
3	Rack tom	Direct out → UMC1820 in 3 (Reaper 3)	10 o'clock
4	Floor tom	Direct out → UMC1820 in 4 (Reaper 4)	2 o'clock
5	Room L	Direct out → UMC1820 in 5 (Reaper 5)	Hard L
6	Room R	Direct out → UMC1820 in 6 (Reaper 6)	Hard R
7	Bass (DI out or mic)	Direct out → UMC1820 in 7 (Reaper 7)	Center
8	Magnatone (SM57)	Direct out → UMC1820 in 8 (Reaper 8)	10 o'clock
9	Marshall or Orange	Direct out → ADA8200 in 1 (Reaper 9)	2 o'clock
10	Lead vocal (Worker Bee)	Direct out → ADA8200 in 2 (Reaper 10)	Center
11	ATW-R3210 wireless	Direct out → ADA8200 in 3 (Reaper 11)	Center
12-13	Reface CS L/R	Direct outs → ADA8200 in 4-5 (Reaper 12-13)	Hard L / hard R
14	Trumpet (SM58)	Direct out → ADA8200 in 6 (Reaper 14)	1 o'clock
15	Vocal 2 (Peavey)	Direct out → ADA8200 in 7 (Reaper 15)	10 o'clock
16	Vocal 3 (Turner 95D)	Direct out → ADA8200 in 8 (Reaper 16)	2 o'clock
17-18, 21	SH-201 L/R, drum machine	**Not recorded** in this layout (monitor only). See "Recording more than 16".	As in the main map
19-20, 23-24	PV6, Tascam 246	Monitor only, same as the main plan	As in the main map

Rewiring: what moves

- **White-tape snake (recording 1-8):** unplug its console end from GROUP OUTS 1-8 and plug the same tails, in the same color order, into the DIRECT OUTS of channels 1-8. The UMC1820 end doesn't move.
- **Blue-tape snake (direct outs to the ADA8200):** move its console tails from the direct outs of channels 10, 11, 14, 15, 16, 17, 18, 21 to the direct outs of channels **9-16**, in color order (brown = ch 9 through black = ch 16). The ADA8200 end doesn't move.
- **Playback does not move.** The black-tape and yellow-tape snakes stay on tape returns 1-16. Reaper track N now plays back on channel N's own tape return, which is how this console was designed to be used.
- Label the console ends with a second strip of tape ("ALT A") so nobody has to guess which layout is plugged in.

Settings that change

- **Channel assign buttons:** on channels 1-16 press **L/R only**, no group buttons. The groups are no longer in the recording path; leave group faders down and the group L-R assign buttons up.
- **Channel faders 1-16 at U (unity), and leave them there while recording.** On the 24•8 the direct out is taken after the EQ, the fader and the mute. Moving a fader changes the recorded level; muting a channel mutes its track. Set the level with the trim knob only, to -12 to -6 dBFS on Reaper's meters.
- **EQ is printed.** Whatever the channel EQ and LO CUT are doing goes onto the track. Keep it to corrective moves (LO CUT on vocals, a small cut where something's boomy) and save the real shaping for Reaper.
- **Inserts still work.** The 76-KT on a channel insert is recorded, because the insert sits ahead of the direct out in the channel.
- **Monitoring:** the hard-panned group issue goes away. With natural pans the control room and MONITOR headphones sound balanced without pressing MONO. Aux 3/4 performer mixes work the same as now.
- **Reaper:** save a second track template, "Toad Studio tracking (direct)", with tracks named in the order of the channel map above (track N = input N = console channel N). Outputs don't change.
- **UMC1820 and ADA8200:** same as now: gain knobs at minimum, UMC inputs 1-2 on LINE, not INST.

Recording more than 16 at once

- This layout records 16 sources, one per track. The main plan squeezes 20 sources into 16 tracks by blending some on the groups. If a session needs the SH-201 or the drum machine, swap a blue snake tail for that session: for example move the tail from ch 15's direct out to ch 17's and from ch 16's to ch 18's, and the SH-201 records instead of vocals 2 and 3.
- Or run a hybrid for that session: put the white snake's last tails back on GROUP OUTS 7-8, assign the SH-201 (ch 17-18) to groups 7-8, and give up the bass and Magnatone tracks (or move them to group pairs too). Hybrids work but are easy to get confused by; write down which tails are where.

Main plan vs Alternate A

	Main plan (groups + direct outs)	Alternate A (all direct outs)
Sources recorded at once	20 sources onto 16 tracks	16 sources, one per track
Drums	Toms and room mics blended onto one stereo pair (groups 3-4); balance fixed at tracking	Every drum mic on its own track; balance decided in the mix
Two guitar amps	Blended onto group 6	Separate tracks (8 and 9)
Noise	Groups add the bus summing stage	Skips the bus; possibly quieter (test it, below)
Recording level set by	Trim, channel fader and group fader	Trim only (faders fixed at U)
Headphone pans	Group channels hard L/R; MONO often needed	Natural pans; no MONO needed
Setup effort	As in the Setup Map	Simpler: L/R assign only on every channel
Best for	Big band takes with everything going	Up to 16 sources when you want to mix later

How the direct outs behave on the 24•8

- **After the EQ, fader and mute.** A direct out on this console is not a raw mic feed. Whatever the channel's EQ, fader and mute are doing is what gets recorded, so faders stay at U and nobody touches mutes during a take.
- **Unbalanced** (1/4" TS). Fine at line level over the 20 ft snakes. Keep the snakes away from power cords; an unbalanced line picks up hum more easily than a balanced one.
- **Independent of the groups and the L/R mix.** The direct out works whether or not the channel is assigned anywhere, so the L/R assign is only for hearing the channel.
- **Noise.** Skipping the groups removes the bus summing stage from the recording path. Groups pick up a little hiss from every channel assigned to them, so a direct out can be quieter. How much depends on the console; the noise test below measures it.

Where the computer comes back in

- **Playback goes to the tape returns, never to channel inputs.** The 24•8 is an in-line console: every channel has a second path (Mix-B) fed by its own tape return. UMC1820 outputs 1-8 feed tape returns 1-8 and the ADA8200's outputs feed tape returns 9-16, exactly as in the main plan. That gives 16 playback tracks while all 24 channel inputs stay free for sources. Plugging playback into channel inputs (for example channels 17-24) would give only 8 playback tracks and take away the channels that hold the SH-201, PV6, drum machine and Tascam.
- In this layout the returns line up one to one: channel N records track N, and channel N's tape return plays track N back. Press MIX-B in the monitor section and raise MIX-B LEVEL to hear playback.
- **The groups don't reach the main mix unless asked.** A group feeds the L/R mix only when its L-R assign button in the master section is pressed. In this layout leave all eight up.

Monitoring and latency

- **Headphones** plug into the console's PHONES jacks (Monitoring chapter). **Speakers** connect to CONTROL ROOM OUTPUTS L/R. Powered bookshelf speakers such as Edifiers need only a cable from 1/4" TRS to their input (RCA or 3.5 mm on most models); turn CNTRL RM down before connecting them.
- **Zero latency for the performer.** The console sends each live mic straight to the headphones and speakers; the computer is not in that path.
- **Backing tracks and earlier takes come from the computer.** They play from Reaper through the tape returns, and new overdubs record into Reaper. Reaper shifts each recorded take back by the interface's latency so it lines up. This works as long as Reaper's input monitoring is off on the audio tracks (Reaper Setup, section 5).

The noise test (do this before deciding)

- Set up one mic, say the kick mic on ch 1, with the trim at its normal recording setting.
- Main-plan path: ch 1 assigned to group 1, group fader at U, recording on Reaper track 1. Record 15 seconds of silence (nobody playing, the mic connected).
- Alternate path: move the white snake's brown tail from GROUP OUT 1 to ch 1's DIRECT OUT, ch 1 assigned to L/R only, fader at U. Record another 15 seconds of silence on the same track.
- In Reaper, solo each clip, turn up the master and listen on headphones, and compare their peak readings (or run each through a plugin meter). A difference of 3 dB or more is real; less than that and the groups weren't the problem, and hiss is more likely coming from trim set too high or a noisy source.
- If the groups do turn out noisier, also try pressing in only the group buttons you need (unassign every unused channel from the groups) in the main plan. That alone lowers bus noise.

Switching back

- Move the white snake back to GROUP OUTS 1-8 and the blue snake back to the direct outs of 10, 11, 14-18 and 21, re-press the group assign buttons from the Setup Map, and load the main Reaper template. Everything else in the book applies again unchanged.

Alternate B: stereo playback (two outputs)

Reaper plays the whole mix back to the console on two outputs instead of sixteen. The console still records all 16 inputs and still handles every live source and headphone mix; the mixing of recorded tracks happens in Reaper. Use it until the 6 ft playback snake is in place, or for good if you prefer to mix in the computer. It works with either the main plan or Alternate A.

Alternate B wiring

• UMC1820 outputs 1-2 (the MAIN OUT jacks) to **tape returns 1-2**, on the brown and red tails of the black-tape snake, which are already connected in the main plan. Nothing else is needed.
• Not used for playback: UMC1820 outputs 3-8, the ADA8200's outputs and the yellow-tape snake. The second optical cable (UMC1820 to ADA8200) is optional; with only one optical cable, set the ADA8200 to its internal clock and the UMC1820 to clock from optical (Setup Map).
• The UMC1820's front MAIN OUT knob controls outputs 1-2: keep it all the way up and use MIX-B LEVEL on the console as the playback volume.

Alternate B settings

• **Reaper:** every track's output goes to the master only; master to outputs 1/2. This is the "Mix playback" mode in Reaper Setup, section 4.
• **Console:** press MIX-B in the monitor section and raise MIX-B LEVEL to hear the playback in the control room and on MONITOR headphones.
• **Performer headphone mixes:** on channels 1 and 2, press SOURCE on the aux 3/4 section so those two aux knobs send the tape return (the Reaper mix) instead of the live channel. Raise them to put the backing track in the aux 3/4 mix. Channels 1 and 2 still record their live mics normally; SOURCE changes only what the aux hears. Or set the phones source to MIX-B plus AUX SEND 3&4.
• **Overdubs:** the same as always. The live source goes through the console to the headphones with no latency, the backing track comes from Reaper, and Reaper's input monitoring stays off on audio tracks.

16-track vs stereo playback

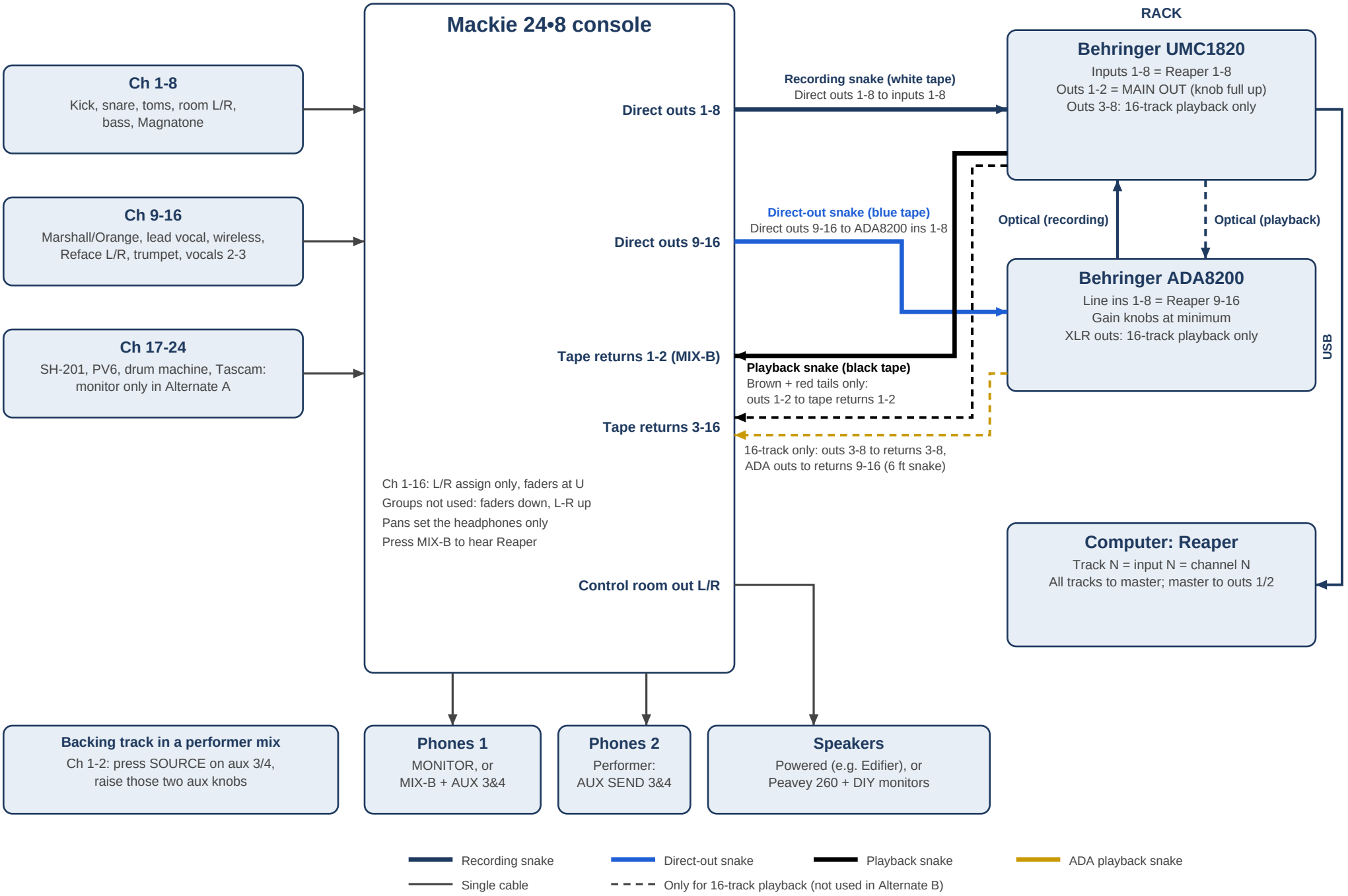
	16-track playback (main plan)	Stereo playback (Alternate B)
Cables	Black and yellow snakes, 2 optical cables	Two tails of the black snake
Where recorded tracks are mixed	On the console: each track on its own tape return, with MIX-B level, EQ and aux sends	In Reaper, with plugins (Reaper Setup, section 7)
TSR-12 and outboard on recorded tracks	Per track, from the console	On the whole mix only, or use Reaper plugins
Backing track in headphones	Any mix of tracks, set per track	The Reaper mix as one stereo feed
Best for	Mixing on the console, analog summing, outboard on playback	Writing, overdubs and mixing in the computer; fewer cables

Moving to 16-track playback later

• Connect UMC1820 outputs 3-8 to tape returns 3-8 (the rest of the black snake), the second optical cable, and the ADA8200's outputs to tape returns 9-16 on the 6 ft snake. In Reaper, send each track to its own output as in the main template. Stereo playback still works at any time by sending tracks to the master only.

Alternate Setups: Connection Diagram

Alternate A recording (all direct outs) with Alternate B playback (two outputs). Dashed lines are only for 16-track playback.



Studio Equipment List

Everything you own. Updated October 2026.

Recording and mixing

Make	Model	Type
Mackie	24•8	8-bus analog console, with meter bridge and 220W power supply
Behringer	U-Phoria UMC1820	Audio interface
Behringer	Ultragain Digital ADA8200	8-channel mic preamp / ADAT expander
Peavey	PV6	Mixer (monitor submixer: e-drums, SamplePad 4, DJ)
Tascam	246 Portastudio	4-track cassette recorder (in repair)
DigiTech	TSR-12	Rack effects processor
Monoprice	8-channel TRS snake (2), 20 ft	Snake cables
—	8-channel snake, 3 ft	Snake cable (swapping for 6 ft)
—	Rolling rack	Equipment rack under the desk
Cockos	Reaper	DAW (desktop PC)

Power

Make	Model	Type
Furman	Merit Series M-8x2	Rackmount power conditioner (won on eBay, on its way)

Monitoring

Make	Model	Type
Peavey	260 Stereo Booster Amplifier	Vintage stereo power amp (for the DIY monitors)
Sony	MDR-7506	Closed-back headphones (used)
Sennheiser	HD 280 Pro	Closed-back headphones (used)

PA / powered mixers

Make	Model	Type
Phonic	Powerpod 410	Powered mixer / PA
Yamaha	EMX62M	Powered mixer

Microphones and wireless

Make	Model	Type
Neat	Worker Bee	Condenser microphone
Shure	SM57	Dynamic microphone
Shure	SM58	Dynamic microphone
—	SM58-style (3)	Dynamic microphones
Peavey	—	Dynamic microphone (came with PA)
Turner	95D	Vintage dynamic microphone
—	—	Snare, kick, and room mics
Audio-Technica	ATW-R3210	UHF diversity wireless receiver
Audio-Technica	ATW-T3201	Body-pack transmitter
Behringer	PS400 Micro Power	Phantom power supply (not needed with the 24•8)

Amplifiers

Make	Model	Type
Magnatone	M10 Custom	Tube guitar amp
Marshall	MG250DFX	2x12 guitar combo amp
Orange	—	Guitar amp
Fender	—	Bass amp

Guitars

Make	Model	Type
Silvertone	ET-230 style (1960s)	Electric guitar
Schecter	— (7-string, white)	Electric guitar
—	—	Acoustic guitar

Keys, drums and electronics

Make	Model	Type
Yamaha	Reface CS	Synthesizer
Roland	SH-201	Synthesizer
Roland	— (1970s)	Drum machine
Roland	—	Electronic drum module + 1 pad
Alesis	SamplePad 4	Percussion pad
—	—	DJ setup
—	—	Drum kit: kick, 14" snare, 13" rack tom, 16" floor tom
Overtone Labs	Tune-bot	Drum tuner

Pedalboard

Make	Model	Type
Electro-Harmonix	720	Stereo looper
Electro-Harmonix	Big Muff	Fuzz
Electro-Harmonix	Cock Fight Plus	Talking wah / fuzz
Death By Audio	Exploding Head	Pedal
—	La Grotte	Pedal
JHS	—	Gain pedal

Not listed yet: the second pedalboard on the amp cart, and the makes and models marked with a dash. Send those and they'll be filled in.

Studio Gear Needs List

Checked against your Studio Equipment List (October 2026): only what's still missing. Low = budget, used, or lowest common price; High = new or bundle price. Tax and shipping not included.

Studio essentials

Item	Suggested model	Qty	Low	High	Notes
Ear pads: Sony MDR-7506	Genuine Sony or aftermarket (Brainwavz, Wicked Cushions, GEVO)	1 pair	$12	$22	Aftermarket around $12 to $17; genuine Sony about $16 to $22.
Ear pads: Sennheiser HD 280 Pro	Genuine Sennheiser pad kit or aftermarket	1 set	$10	$25	Aftermarket pairs around $10 to $15. Genuine kit (ear + headband pads) about $24, fits 2016 and newer units.
XLR cables	15-20 ft for drums, amps and trumpet; 25 ft for vocals and room mics	Up to 15	$90	$150	Estimate for budget cables. The channel map uses 14 mic lines plus 1 spare; subtract any you already own.
Optical (TOSLINK) cable	Any standard TOSLINK cable, 3-6 ft	1	$8	$15	Estimate. UMC1820 ADAT out to ADA8200 ADAT in, for 16-track playback and clock. Add one more if you don't have the ADA8200-to-UMC1820 cable yet.
Third 8-channel TRS snake	Monoprice (to match yours) or Hosa CSS-801 / CSS-803	1	$43	$56	Direct outs from ch 10-11, 14-18 and 21 to the ADA8200 (all 8 tails). Eight single TRS cables also work. The two Hosa snakes in Cory's listing ($20 each) could cover this.
Subtotal			**$163**	**$268**	

For the Tame Impala sound

Item	Suggested model	Qty	Low	High	Notes
FET compressor	Klark Teknik 76-KT	1	$135	$209	$209 new; around $135 used. 1176-style, for squashed drums and vocals. Two dbx 1066s from Cory's listing ($150 each) are the alternative.
Insert (Y) cables	TRS to dual TS insert cable	2	$20	$40	Estimate. One connects the 76-KT to a channel insert on the 24•8; the second is a spare. Four channels of compression would need 4; add 4 more for the gate.
Vibrato pedal	Diamond Vibrato (reissue)	1	$249	$279	About $249 to $279 new. The pedal Parker used on most Lonerism guitars. Needs a 9-18V supply; check whether one is included.
Phaser pedal	MXR Phase 90	1	$100	$100	$99.99 new; often cheaper used. Needs a 9V adapter or battery.
Subtotal			**$504**	**$628**	

Monitoring and recording extras

Item	Suggested model	Qty	Low	High	Notes
DIY passive monitor kit (pair)	Parts Express C-Note MT kit, or the Hitmaker MT studio monitor kit	1 pair	$160	$250	C-Note pair with knock-down cabinets about $159.98; drivers, crossovers and cabinets included. Upper figure is an estimate for the Hitmaker. Check the impedance against the Peavey 260's rating.
Peavey 260 tech check	Local amp tech	1	$60	$150	Rough estimate: clean controls, check capacitors, fit a grounded 3-prong plug if it has a 2-prong.
Line cables to the Peavey 260	1/4" TRS (console) to 1/4" TS (amp)	2	$15	$30	Estimate. CONTROL ROOM out L/R to Peavey channels A and B.
Speaker cable	Real speaker cable, length to suit	2	$15	$30	Estimate. Peavey outputs to the DIY monitors. Never instrument cables.
DI box	Radial ProDI	1	$80	$129	About $80 used, around $129 new. Passive, with a -15dB pad and ground lift. Clean guitar/bass track for re-amping; acoustic with a pickup.
Rack shelf for the computer	2U vented rack shelf (on order)	1	$20	$40	Estimate. Holds the PC in the rack above the Furman. A lip or strap so it can't slide when the rack rolls.
Monitor swing arm	Desk-clamp monitor arm, VESA	1	$25	$60	Estimate. Clamps to the desk right of the console; keeps the screen off the console surface.
Speaker clamp mounts	Pair of desk-edge clamp speaker brackets, or a DIY wood bracket	1 pair	$20	$60	Estimate. Mount the DIY monitors on the back edge of the desk, tweeters at ear height.
Reference cable	3.5mm to dual RCA	1	$5	$12	Estimate. Phone or computer into the 2-TK input to A/B against reference tracks.
Subtotal			**$400**	**$761**	

Budget DI alternative: Behringer DI400P, about $30 new. Pick one DI or the other; totals use the Radial ProDI.

Only if needed

Item	Suggested model	Qty	Low	High	Notes
Delay / second effects unit	Used rack delay or multi-effects, e.g. TC Electronic D-Two, Boss RV/DD rack units, or a second TSR-12	1	$60	$200	Estimate. Aux send 2 to its input, outputs to aux return 2. Takes the rack slot the headphone amp would have used.
Personal headphone mixers	Small 4-channel mixers with a headphone jack (Behringer Xenyx 502 class), one per extra player	1-3	$30	$60	Estimate, each. Fed from aux sends 3-6; each one doubles as that player's headphone amp. Replaces a rack headphone amp.
XLR-female to TRS-male adapters	Any brand	8	$15	$30	Estimate. Only if the 6 ft snake from Todd is TRS on both ends (ADA8200 outputs are XLR).
Subtotal			**$105**	**$290**	

Compressor alternatives and gate option (not in the totals)

Decision: the 76-KT is the primary compressor and is in the totals above. These are the alternatives seen on Facebook Marketplace, kept here in case a 76-KT doesn't turn up or you want four channels later. Prices are the listed asking prices.

Item	Suggested model	Qty	Low	High	Notes
Alternative B: two dbx 266XS	Drew's Facebook listing	1 pair	$110	$110	Four channels of clean compression plus a gate on each, for less than one 1066. Covers the drum pair (linked), lead vocal and bass. Doesn't give the 1176 crunch; the 76-KT or ReaComp still does that. Test every channel and gain-reduction meter.
Alternative C: two dbx 1066	Cory's Facebook listing	1 pair	$300	$300	Same four channels, better built. Only worth it over the 266XS pair if the price comes down.
Quad noise gate	Klark Teknik DN514+ (Facebook, $100, free shipping)	1	$100	$100	Fair price. Four channels on the inserts of kick, snare and both toms; cuts ring and cymbal bleed before it's recorded. Reaper can gate for free afterwards, so this is a convenience, not a must. Needs 4 insert Y cables.
Subtotal			**$510**	**$510**	

Totals

Estimated total	Low	High
Studio essentials	$163	$268
Essentials + Tame Impala sound	$667	$896
Essentials + Tame Impala + monitoring extras	$1,067	$1,657
Everything, including only-if-needed items	**$1,172**	**$1,947**

The Goal Studio Diagram on the next page shows where every item above plugs in once it's bought.

What's not on the list, and why

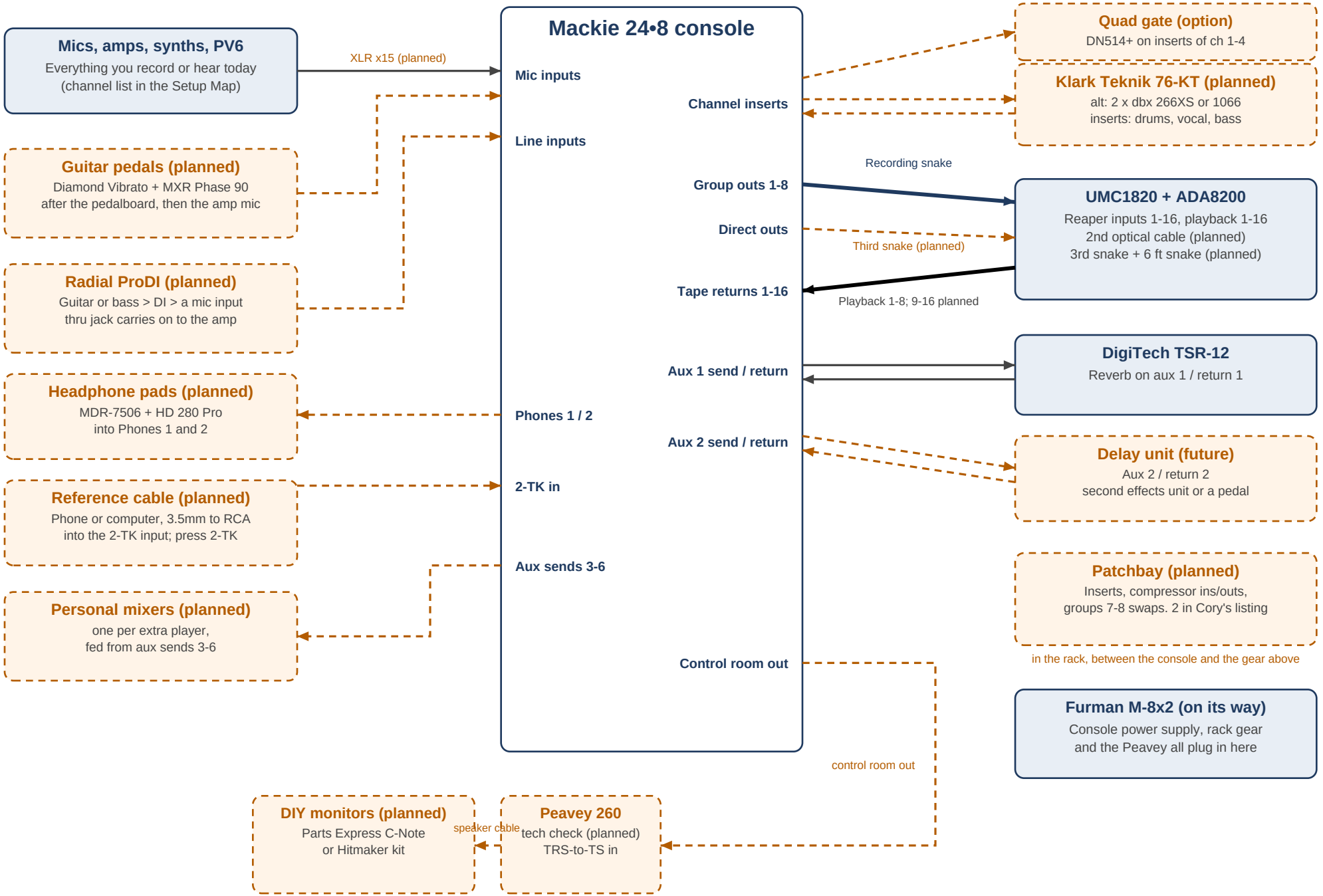
Not on the list	Why
Audio interface	You have the Behringer UMC1820, plus the ADA8200 for 8 more inputs.
Second 8-channel snake	You have two 20 ft snakes: one for recording, one for playback.
Furman M-8x2 power conditioner	Won on eBay; on its way.
Rack	You have the rolling rack under the desk.
Power amp (Crown)	Not needed: the Peavey 260 stereo power amp drives the DIY passive monitors.
Powered monitors (JBL 305P MkII)	Replaced by the Peavey 260 + DIY passive monitors. Still the plug-and-play backup: $169 each new, about $80-124 used.
Graphic EQ (dbx 231)	For correcting PA speakers and rooms, not studio recording; console EQ and ReaEQ cover it. Maybe for the garage PA.
Mic preamp	Not needed: the 24•8 has 24 preamps, plus 8 on the ADA8200.
More microphones, SM57s	Every mic position is covered (SM57, SM58, 3 SM58-style, Peavey, Turner 95D, Worker Bee, drum and room mics). Another SM57 is the first add if you want one.
AT2020 condenser	You have the Neat Worker Bee.
Lexicon MX200 reverb	Your DigiTech TSR-12 covers reverb and delay on an aux send.
dbx 166XS compressor	Replaced by the 76-KT, which suits the drum sound better. Reaper's free ReaComp covers the mix bus.
Mic stands	Add stands only if you're short.

Not on the list	Why
Patchbay and patch cables	Nice to have, not needed to get started. Two are in Cory's listing at $20 each.
Headphones	A used Sony MDR-7506 and Sennheiser HD 280 Pro at $40 each. Only replacement pads are listed.
Rack headphone amp (HA8000)	Dropped to keep the rack slot for a delay unit. Personal mixers at the players' spots do the job and don't need rack space.
Fuzz Face, pedal power supply	The Big Muff on your pedalboard covers fuzz. The Diamond Vibrato and Phase 90 still need their own supplies.

Notes: Prices come from major retailers and used-market listings and change often. Items marked "Estimate" were not priced from a specific listing. The Tascam 246 repair isn't included. The Behringer PS400 isn't needed with the 24•8, which has built-in phantom power.

Goal Studio: Everything on the Needs List Connected

Where each planned purchase plugs in. Orange dashed = on the needs list. Navy solid = already in the studio.



Studio Equipment Values

Rough used-market estimates, October 2026. Low = quick private sale or rough condition; High = good condition, patient sale. The computer and the second pedalboard aren't included.

Console, rack and recording

Item	Low	High	Notes
Mackie 24•8 with meter bridge and 220W power supply	$600	$1,200	A refurbished 24x8x2 + power supply bundle was listed at $1,199.99 by a vintage dealer; private sales run lower.
Behringer UMC1820	$150	$200	
Behringer ADA8200	$150	$200	
DigiTech TSR-12	$50	$100	
Furman M-8x2	$60	$90	Won on eBay.
Peavey PV6	$40	$80	
Tascam 246 Portastudio (not playing tape)	$75	$150	Worth noticeably more once repaired.
Snakes: 2 x 20 ft TRS, 1 x 3 ft	$70	$100	
Rolling rack	$50	$100	
Subtotal	**$1,245**	**$2,220**	

Amplifiers

Item	Low	High	Notes
Magnatone M10 Custom	$600	$1,100	Equipboard average about $740; a 1965 M10A in excellent working condition was listed at $1,349.99.
Marshall MG250DFX (2x12 combo)	$200	$300	Solid-state stereo combo.
Fender bass amp and Orange guitar amp	$250	$900	Models unknown; the biggest swing in this category.
Peavey 260 Stereo Booster Amplifier (1970s)	$75	$175	Rough estimate; untested vintage power amp.
Subtotal	**$1,125**	**$2,475**	

Guitars

Item	Low	High	Notes
Silvertone ET-230 style (1960s, likely Teisco-made)	$300	$800	Teisco built guitars for Sears' Silvertone line; the ET-230 is the 1960s Teisco "sharkfin" design. Originality and condition drive the price.
Schecter 7-string (white)	$200	$350	Exact model unknown.
Acoustic guitar	$100	$400	Brand and model unknown.
Subtotal	**$600**	**$1,550**	

Synths, drums and electronics

Item	Low	High	Notes
Yamaha Reface CS	$250	$330	
Roland SH-201	$350	$500	Equipboard average about $395; a used one from Japan listed on eBay at $494.
1970s Roland drum machine	$300	$2,000	Model unknown. A CR-78 is the high end.
Roland e-drum module + 1 pad	$50	$150	
Alesis SamplePad 4	$80	$120	
DJ setup	$200	$800	Brands and models unknown.
Drum kit (kick, 14" snare, 13" and 16" toms)	$200	$500	
Overtone Labs Tune-bot	$50	$80	
Subtotal	**$1,480**	**$4,480**	

Microphones, wireless and headphones

Item	Low	High	Notes
Neat Worker Bee	$80	$120	
Audio-Technica ATW-R3210 + ATW-T3201	$150	$250	
Shure SM57	$60	$75	
Shure SM58	$60	$75	
SM58-style mics (3)	$30	$60	
Peavey mic	$15	$30	
Turner 95D (vintage)	$50	$150	
Kick, snare and room mics	$50	$150	Models unknown.
Behringer PS400	$15	$25	
Sony MDR-7506 + Sennheiser HD 280 Pro (used)	$80	$100	Priced at $40 each; both need pads.
Subtotal	**$590**	**$1,035**	

Pedalboard

Item	Low	High	Notes
Electro-Harmonix 720 looper	$100	$140	
Electro-Harmonix Big Muff	$50	$80	
Electro-Harmonix Cock Fight Plus	$100	$140	
Death By Audio Exploding Head	$120	$200	
La Grotte	$100	$200	
JHS gain pedal	$100	$180	
Subtotal	**$570**	**$940**	

Powered PA mixers

Item	Low	High	Notes
Phonic Powerpod 410	$100	$200	
Yamaha EMX62M	$100	$180	
Subtotal	**$200**	**$380**	

Totals

Category	Low	High
Console, rack and recording	$1,245	$2,220
Amplifiers	$1,125	$2,475
Guitars	$600	$1,550
Synths, drums and electronics	$1,480	$4,480
Microphones, wireless and headphones	$590	$1,035
Pedalboard	$570	$940
Powered PA mixers	$200	$380
Total	**$5,810**	**$13,080**

Most likely value: around the middle of the range. The widest swings come from the drum machine model, the Fender and Orange amp models, the Silvertone, the DJ setup and the acoustic guitar. Replacing everything new would cost much more, probably $12,000-20,000+.

For insurance: photograph each piece and record serial numbers. Researched prices (Mackie, Magnatone, SH-201) and the Silvertone/Teisco background come from dealer listings, eBay and Equipboard; all other figures are rough estimates, and prices change often.

How Each Piece Works

What each piece of your gear does and how it connects to the Mackie 24•8. For exact channel numbers and cables, see the Studio Setup Map.

How it all connects

Signal	From	To
Mics and instruments	Mics (XLR), synths and mixers (1/4")	Console mic and line inputs
Recording 1-8	Group outs 1-8	UMC1820 inputs 1-8 (white-tape snake)
Recording 9-16	Direct outs of ch 10-11, 14-18, 21	ADA8200 line ins 1-8 (blue-tape snake), then optical to the UMC1820
Playback 1-8	UMC1820 outputs 1-8	Tape returns 1-8 (black-tape snake)
Playback 9-16	ADA8200 outputs 1-8	Tape returns 9-16 (yellow-tape snake)
Your monitoring	Control room outs L/R; Phones 1 and 2	Peavey 260 and DIY monitors; headphones
Performer mix	Aux sends 3-6 (PRE down)	Phones 2, or personal mixers at each player
Effects	Aux send 1 (PRE up)	TSR-12 input; its outputs to aux return 1
Dynamics	Channel insert (Y cable)	Compressor in and out
Tape	Main mix out L/R	Tascam 246 line inputs; its line outs to ch 23-24
Power	Wall outlet	Furman M-8x2, then all gear

Recording

Audio interface Behringer UMC1820	
Purpose	Turns the console's analog signals into digital audio so you can record, edit and mix in Reaper, then sends audio back out to the board.
How it works	Eight analog inputs (1-2 on the front, 3-8 on the back) and ten analog outputs. Outputs 1-2 are the MAIN OUT jacks, which follow the front MAIN OUT knob. It connects to the computer by USB and has optical (ADAT) in and out plus MIDI in and out.
Setup with the 24•8	Group outs 1-8 go to inputs 1-8. Set the inputs to LINE with gain at minimum and do the gain staging on the console. Outputs 1-8 go to tape returns 1-8 so Reaper playback comes back through the board; keep the MAIN OUT knob all the way up.

Input/output expander Behringer ADA8200	
Purpose	Adds eight more inputs and eight more outputs to the UMC1820 over optical cables, for 16 tracks in and 16 out.
How it works	Eight mic/line inputs on the front and eight XLR outputs on the back. One optical cable carries its 8 inputs to the UMC1820; a second carries 8 Reaper outputs back to it. ADAT carries 8 channels at 44.1 or 48 kHz and only 4 at 88.2 or 96 kHz.
Setup with the 24•8	Direct outs from channels 10, 11, 14-18 and 21 feed its line inputs with the gain knobs at minimum. Its XLR outputs feed tape returns 9-16. The two units have to share one clock: see the Setup Map for the SYNC settings.

8-channel snakes 2 x 20 ft TRS owned; a third TRS and a 6 ft snake to come	
Purpose	Carry eight line-level signals in one bundled cable, which keeps the back of the console organized.
How it works	Eight color-coded tails on each end. Yours aren't numbered, so the color order is the numbering: brown, red, yellow, green, blue, gray, white, black.
Setup with the 24•8	One snake per job, marked with tape near both ends: white for recording, black for playback 1-8, blue for direct outs, yellow for ADA8200 playback 9-16. Coil extra length loosely and keep it away from power cords and the console's power supply.

Monitoring

Power amp and monitors Peavey 260 Stereo Booster Amplifier + DIY passive speakers	
Purpose	Give you speakers to mix on, so mixes translate to cars, earbuds and other systems. Right now all mixing is on headphones.
How it works	The Peavey is a 1970s stereo power amp: it makes a line-level signal strong enough to drive passive speakers. It has no preamp or tone controls. The speakers come from a kit (drivers, crossovers and cabinets).
Setup with the 24•8	CONTROL ROOM out L/R to the Peavey's channel A and B inputs, then real speaker cable to the speakers. Have a tech check the amp first, and match the speakers' impedance to the minimum on its back panel. The CNTRL RM knob is your volume. Place the speakers and your head in an equal-sided triangle with the tweeters at ear height.

Closed-back headphones Sony MDR-7506 and Sennheiser HD 280 Pro	
Purpose	Let people hear the mix while tracking without the sound leaking into open mics.
How it works	Closed-back cups seal around the ear to block outside noise and keep sound in.
Setup with the 24•8	The console has two headphone outputs, PHONES 1 and PHONES 2, each with its own level knob and source buttons. Press MONITOR to hear the control room mix, or AUX SEND 3/4 for a separate performer mix.

Personal headphone mixers (planned) Small mixers with a headphone jack, one per extra player	
Purpose	Give each extra musician their own headphone volume and balance, without a rack headphone amp.
How it works	A small mixer takes two or three aux sends from the console, each carrying a group of instruments, and the player blends them at their own spot and plugs headphones into the mixer.
Setup with the 24•8	Feed them from AUX SENDS 3-6 with PRE down, so the mixes don't change when you move channel faders. Up to two people still use the console's own PHONES 1 and 2.

Microphones

Dynamic mics Shure SM57, SM58, three SM58-style, Peavey, Turner 95D	
Purpose	Rugged mics for loud sources: drums, guitar amps, trumpet and live-style vocals.
How it works	Dynamic mics generate their own signal and don't need phantom power. They handle very loud sounds and pick up mostly from the front, which reduces bleed.
Setup with the 24•8	XLR into a channel mic input. For guitar amps, start an inch or two from the grille, aimed near the center of the speaker cone. The Turner is a vintage mic with its own connector and may be high impedance: check it before plugging in (see the Setup Map).

Condenser mic Neat Worker Bee	
Purpose	Captures more detail and air than a dynamic mic: lead vocals and acoustic guitar.
How it works	Condenser mics need +48V phantom power from the console. They're more sensitive than dynamics, so they pick up quieter sounds and more of the room.
Setup with the 24•8	XLR into channel 10, then switch on phantom power for channels 9-16. Turn the faders and monitors down before switching phantom on or off to avoid a loud pop. Record in a quiet, less echoey spot and use a pop filter for vocals.

Wireless system Audio-Technica ATW-R3210 receiver + ATW-T3201 body pack	
Purpose	Carries a mic or instrument to the console without a cable.
How it works	The body pack transmits to the receiver, which puts out a normal balanced signal.
Setup with the 24•8	XLR from the receiver's output to channel 11. Set the receiver's output level before setting the channel trim.

DI box (on the needs list) Radial ProDI	
Purpose	Lets a guitar, bass or acoustic pickup plug straight into a console mic input.
How it works	A transformer converts the instrument's weak, high-impedance signal to a balanced mic-level signal. A thru jack passes the instrument on to the amp, and a ground lift switch cures hum.
Setup with the 24•8	Instrument into the DI, thru jack to the amp, XLR out to a mic input. You get a clean track alongside the amp mic that can be re-amped later. Synths and drum machines don't need one.

Effects and processing

Reverb / multi-effects DigiTech TSR-12	
Purpose	Adds space and depth (reverb, delay) so dry sounds don't feel flat in the headphones.
How it works	A rack effects processor with stereo outputs. You pick a program and adjust it from the front panel.
Setup with the 24•8	AUX SEND 1 to its left input; its L/R outputs to AUX RETURN 1. Set its mix to 100% wet, since the console blends in the dry signal. Turn up aux 1 (PRE up) on each channel that needs the effect. It's in the monitor path only, so Reaper records dry.

Compressor (on the needs list) Klark Teknik 76-KT, or dbx 1066	
Purpose	Evens out volume swings so vocals, bass and drums sit steady, and gives drums the squashed 70s sound.
How it works	Above the threshold it turns the signal down by the ratio; attack and release set how fast it reacts; output gain restores the level. The 76-KT is a single-channel, aggressive 1176-style unit. The dbx 1066 is two cleaner channels with gates.
Setup with the 24•8	Plug it into a channel's insert jack with an insert (Y) cable: the TRS end goes into the insert and the two other ends go to the compressor's input and output. On Mackie inserts, the tip is the send and the ring is the return. What it does is recorded.

Patchbay (optional) Two are in Cory's listing	
Purpose	Puts key connections on one front panel so you can reroute with short patch cables instead of reaching behind the console.
How it works	Each channel has a top and bottom jack front and back. Normalled pairs stay connected until you plug in at the front.
Setup with the 24•8	Start small: channel inserts, the compressor's ins and outs, and the connections you swap most (groups 7-8 for the Reface or the Tascam). Label every point.

Other gear

Monitor submixer Peavey PV6	
Purpose	Combines sources that only need to be heard (e-drum module, SamplePad 4, DJ mixer) into one stereo pair, saving console channels.
How it works	A small mixer: each source gets a level control, and the main out carries the blend.
Setup with the 24•8	Main out to channels 19-20, panned left and right, L/R button only. Keep its gains and master at unity. Don't put anything through it that you want on its own track.

4-track cassette Tascam 246 Portastudio (in repair)	
Purpose	Analog tape recording and tape saturation.
How it works	Records four tracks to cassette and has its own small mixer.
Setup with the 24•8	MAIN MIX out to two of its line inputs to record to tape; its stereo line out to channels 23-24 to hear it. Keep 23-24 out of the L/R mix while recording to tape. To move tape into Reaper, assign 23-24 to groups 7-8.

Power conditioner Furman M-8x2	
Purpose	Protects gear from surges and filters electrical noise that can show up as hum or buzz.
How it works	A rackmount power strip with surge suppression and filtering. Its switch turns the whole rack on or off.
Setup with the 24•8	It sits just above the Mackie power supply with one empty space between. Plug the power supply, UMC1820, ADA8200 and outboard gear into it. Running everything from one conditioner on one circuit also helps prevent ground-loop hum.

Rack Rolling rack under the desk	
Purpose	Holds the rackmount gear and the console power supply in one ventilated place.
How it works	Standard 19" rails measured in rack units (1U = 1.75").
Setup with the 24•8	Top to bottom: UMC1820, ADA8200, TSR-12, 76-KT, delay unit, a shelf with the computer, Furman, Mackie power supply. The power supply goes at the bottom because it's heavy, runs warm and can put hum into nearby audio gear; the computer's shelf sits above it with air space on both sides.

Power-up and power-down

Power-up order	Power-down order
1. All levels down	1. Save; all levels down
2. Furman switch: console PSU, UMC1820, ADA8200, TSR-12, compressor, delay all come up	2. Peavey 260 off
3. Computer, then Reaper	3. Close Reaper, shut the computer down
4. Wait 20-30 seconds	4. Furman switch off (everything on it goes down together)
5. Peavey 260, on its own switch	5. Leave the Furman plugged into the wall
6. Bring levels up: CNTRL RM, Peavey, phones, main fader	

The speaker amp goes on last and off first so turn-on thumps from other gear never reach the speakers. The Peavey and the computer get their own wall outlets for that reason; everything else is on the Furman. Outlet-by-outlet detail is on the Power page.

Quick gain-staging checklist

- Trim down, fader at U, channel unmuted.
- Have the performer play or sing at full volume. Press SOLO and raise the trim until peaks regularly reach 0 on the L/R meters.
- Leave channel faders and group masters at U and check that Reaper peaks around -12 to -6 dBFS. Fix levels with the trim.
- Leave headroom: tape-style "hot" levels aren't needed when recording digitally.
- The control room and phones knobs set listening volume; they don't affect what gets recorded.
- The full version is in the Gain Staging Guide.